5DATA004C – Data Science Project Lifecycle (2025/26)

Final Project Report

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GitHub Repository: https://github.com/Yalin023/Hotel-Chain-A-Booking-Cancellation-Analysis

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1. Executive Summary

For Hotel Chain A, there is a major business issue involving booking failures in the form of cancellation and no shows among their hotels categorized into City, Airport, and Resort hotels. This impacts their occupancy rates, makes demand prediction difficult, and results in preventable losses of revenue because room sales for hotels are very perishable. This is why this particular project was designed to look at the reasons behind booking failures, determine the financial impact, and develop predictive methods.

The project involved a complete data science methodology that included business understanding, project planning, data preprocessing, exploratory analysis, loss-revenue prediction, predictive modeling, and operation recommendations. The analysis was conducted using the training, validation, and testing datasets provided by the client alongside the data dictionary. In the course of data preprocessing, quality problems like duplicate reservation ID numbers, conflicting reservation status labels, and erroneous dates were found and fixed. Moreover, several new features, such as lead time, length of stay, family size, modified room rates, and expected booking revenue, were generated to enhance interpretability and modeling value.

The process covered the entire data science approach consisting of understanding the business needs, planning the project, data preparation, exploratory analysis, forecasting losses/expected income, predictive modeling, and operationalization recommendations. The exploratory analysis was done based on the provided training, validation, and test datasets, together with a data dictionary. During data preparation, issues related to poor quality of data occurred, namely duplication of reservation IDs, inconsistent statuses for reservations, and incorrect dates. What is more, additional variables, including lead time, length of stay, family size, room rates adjustments, and expected booking income, were calculated.

The revenue lost due to failure of bookings proved the significant financial consequences of such failures. Loss in revenues associated with cancellations was estimated at 1,160,410.45, with losses from no shows amounting to 594,017.75. Consequently, total revenue lost for failed stays was

equal to 1,754,428.20. Region-wise analysis demonstrated that the South region made the biggest contribution to loss due to cancellations, while resort hotels had the highest revenues exposed to risk. Further, segment-level analysis found that high-risk segments were those located mostly in South regions and online reservations.

In an effort to promote forward-thinking decision-making, three classification techniques were compared: Logistic Regression, Decision Tree, and Random Forest. With regard to current validation findings, the best business-driven performance was attained from the use of the Decision Tree, since it produced the best recall of failed bookings, thus emerging as the current best option for detecting bookings that might fail due to cancellations or no-shows. The current findings also indicate that current prediction performance is relatively modest, implying that the predictive model is more applicable as a decision support tool rather than being adopted as an operational tool in its current form. In terms of feature importance, predictors such as meal category, month of booking, net room price, and age were found to drive prediction performance.

The report thus suggests that Hotel Chain A should pursue a risk management strategy with regard to its hotel reservations based on the above findings. The focus of the effort should be on medium to high risk reservations particularly those which are part of segments where losses have been substantial such as Resort and South region. Some of the recommended activities in the operations domain are sending reminders, segment-specific confirmation of reservations, segment-specific analysis of deposit and cancellation policies, and gathering additional data fields such as reasons for cancellation, loyalty status, payment behavior, and changes made in the reservation process.

Generally speaking, this project shows that data regarding reservation history can be a source of valuable information about the financial implications and prediction models related to booking failures of Hotel Chain A. Although the model needs more fine-tuning before making claims about its modeling performance, it still serves a purpose of building more reliable grounds for revenue protection and booking management.

2. Background & Problem Domain

2.1. Organisational background

Hotel Chain A has a wide hospitality portfolio covering City Hotels, Airport Hotels and Resort Hotels with a wide range of business, transit and leisure travelers. The strategic direction of the organisation focuses on enhancing revenue efficiency and customer satisfaction by using data to make decisions. In that environment, minimizing unnecessary losses associated with booking cancellations and no-shows will not be a single operation problem, but a broader attempt to enhance demand forecasting, pricing approach, and operational planning within various types of hotels.

This business environment renders the issue especially significant to analytical study. Hotel operation is characterised by a more or less fixed short-term capacity, perishable inventory, demand time variation and a high degree of customer price sensitivity. Within the concept of revenue management, an unsold room night cannot be stocked and resold in the future, thus, the preventable booking failure involves a direct threat to the transformation of demand into realised revenue. These situations are precisely the type of environment where revenue management and predictive analytics can be employed to generate value.

2.2. Industry context and problem domain

Reservations in hospitality are not just customer records but future indicators to be used in planning occupancy, prices, labour, and service preparations. Hotel forecasting is the continuous process of predicting future room demand and hotel revenues and aids operational, financial, and business decisions (staffing, budgeting, pricing policy, and inventory control). The quality of the downstream decisions is undermined when the outcomes of reservations become unreliable due to cancellations and no-shows.

No-shows and cancellations are not administrative inconveniences. They are closely associated with more general revenue-management issues including underutilized capacity, pressure of overbooking, compensation risk and disturbance to customer service. Previous literature on

hospitality revenue management has demonstrated that late cancellations and no-shows can decrease the potential revenue of perishable room inventory and compel hotels to tradeoff between occupancy protection and the risk of overselling.

In the case of Hotel Chain A, the problem domain is further complicated by the reality that the results of the reservation process are not caused by one effect, but a combination of numerous ones. The project brief singles out customer demographics, booking behavior, lead times, seasonality, the type of hotel and pricing-related variables as potential causes of cancellation risk. It further clarifies that the present method is to a large extent retrospective and this hinders the organisations capability of intervening in time and controlling the risk proactively.

2.3. Business problem

The main business issue is that Hotel Chain A suffers a huge loss of revenues due to booking cancellations and no-shows, but it does not have an active mechanism that allows recognizing high-risk reservations prior to the anticipated day of stay. Practically, this translates to the exposure of the organisation to a number of related weaknesses, which include direct loss of room-revenue, reduced forecast reliability, inefficient pricing and allocation decisions, and uncertainty in operations of its various types of hotels.

It is not merely the question to quantify the number of reservations that are not made. The better business question is what reservations the highest probability of failure has, why they fail and what customer or booking segments cause the greatest loss of revenue. When presented in such a manner, it becomes a decision-support problem as opposed to an entirely descriptive reporting task. It is not just to provide explanations regarding past losses, but to transform historical booking data into predictive and actionable intelligence that can be used to support intervention.

2.4. Why this problem matters

This issue is relevant since hotel income is tied to the process of translating the booked room nights into the final stays. Through cancellations or failure to show up by a guest, the hotel not only loses

its projected room revenue but also planning certainty. Its effects are felt in the pricing, room distribution, personnel, and timing of promotion. This is particularly significant in a portfolio like that of Hotel Chain A whereby the city, airport and resort property would tend to share varied demand trends, reservation periods and customer trends.

Strategically, it is more purposeful to have a predictive ability than a retrospective reporting. Recent studies on hotel-cancellation indicate that individual-level prediction of cancellations can be used to guide managerial decisions on pricing, booking controls, and other policies related to the operations of the business, particularly when the model is interpretable to the business users. This is significant since a model that is good in terms of statistics but cannot be translated to managers would have a less chance of affecting actual decisions.

2.5. Stakeholders and decision context

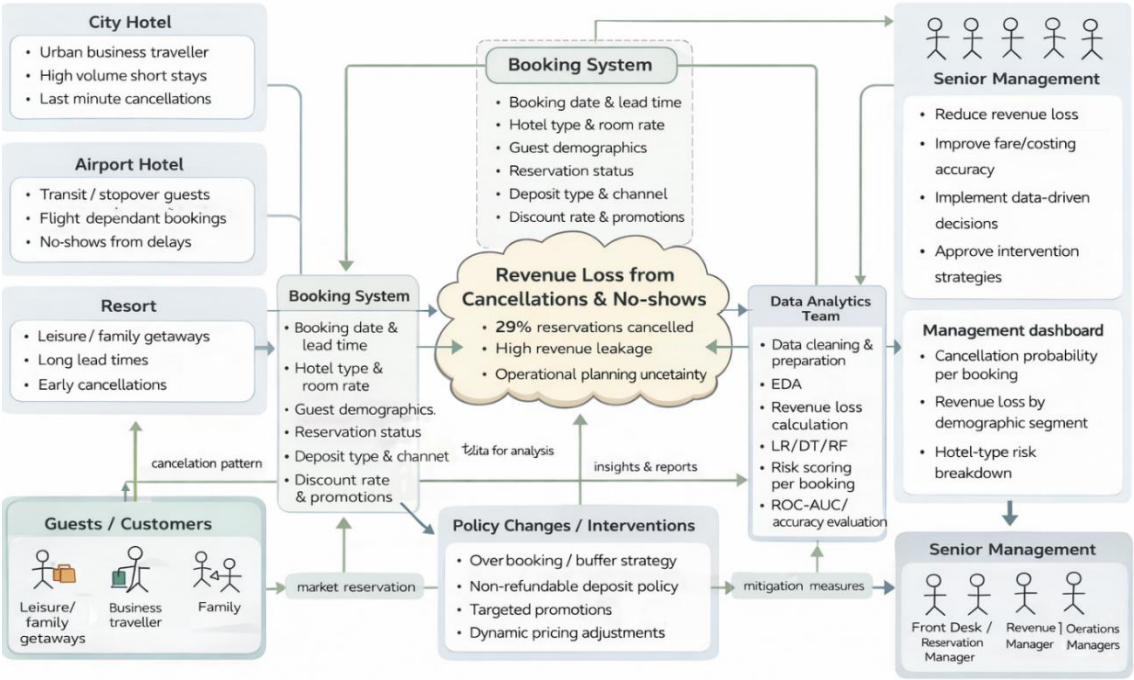
This problem affects several groups of stakeholders. To make investment and policy decisions, the senior management must have a clear picture of the magnitude and the allocation of revenue loss. Revenue managers need to have trustworthy information of booking risk in order to make pricing, deposit and availability decisions. Better forecasts enable operations managers to plan the staffing, housekeeping and service readiness. The analytics capability requires a scalable and repeatable workflow capable of converting raw reservation data into business-understandable information. Indirect stakeholders are also customers, as incorrect forecasting and reactive overbooking reactions may have an impact on the quality of services and the sense of rightfulness.

Due to the various priorities held by these stakeholders, the issue cannot be narrowed to one measure like accuracy. The analytical solution should be a compromise between predictive performance and transparency, operational usefulness and alignment with commercial decision-making. This renders the problem technical and organisational in nature.

2.6. Problem structuring perspective

Booking cancellations at Hotel Chain A are the right way of thinking that involves a systems viewpoint as they are the interactions between the customers, reservation rules, pricing policies, operational procedures, and external demand factors that cause the cancellation. The problem in the project context is vague, as the business is aware of losses happening, but not entirely aware of the mechanisms that drive them; complex, as it involves many stakeholders, types of hotels, and characteristics of bookings; and uncertain since at the time of booking it is not known with any certainty whether the booking will lead to losses. Such situations are the ones where problem-structuring methods come in handy.

Soft Systems Methodology will be particularly appropriate in this case since it promotes exploring the problem with rich pictures, root definition, and CATWOE analysis and then proceeds to technical modelling. This gives the report strength in that it demonstrates that the project is supported by business knowledge as opposed to plunging straight into code and visualisations.



2.7. Root definition

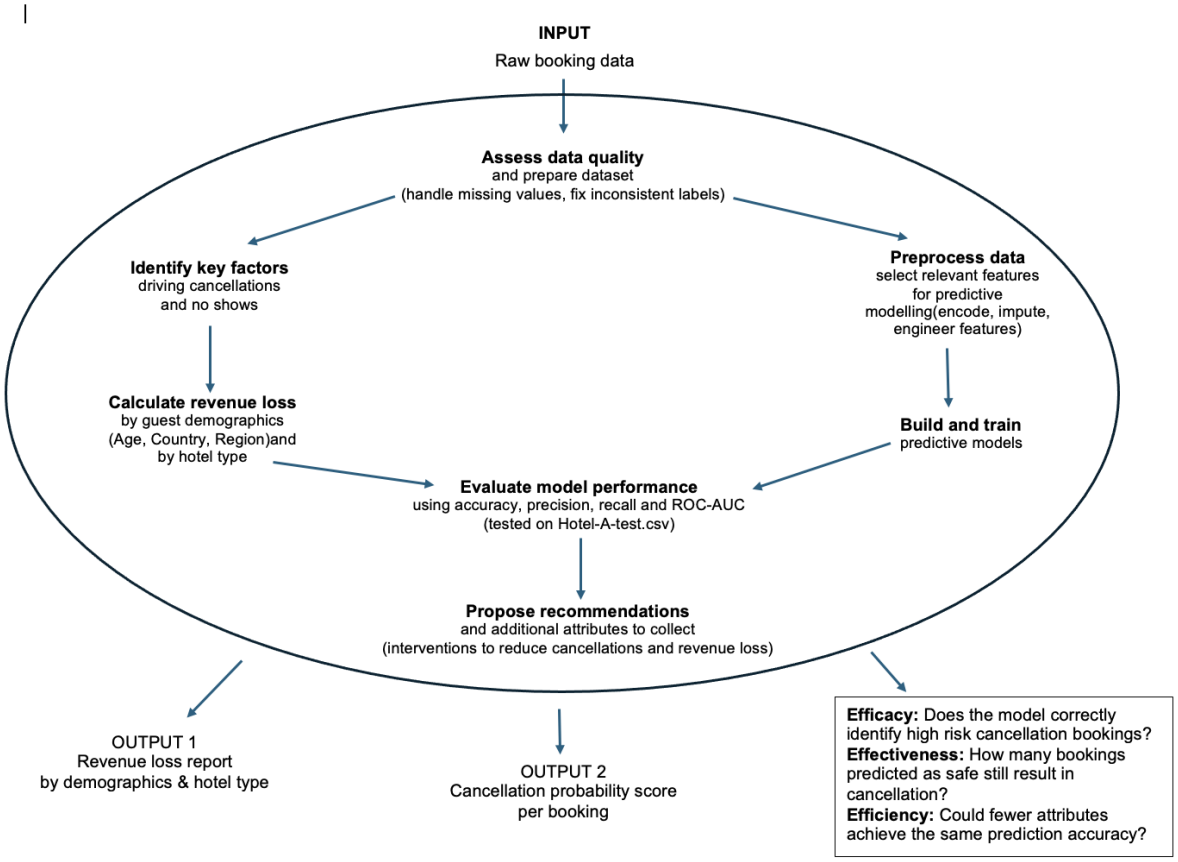
In order to minimize preventable revenue loss, enhance forecasting and booking management, and facilitate more proactive operational and commercial decision-making within the limitations of data quality, interpretability, and business practicality, Hotel Chain A owns a data-driven decision-support system that its management and analytics teams use to convert historical booking and customer data into actionable insight on cancellation and no-show risk.

2.8. CATWOE analysis

C Client *(Who benefits / is affected?)*	Hotel Chain A management and hotel staff who are losing revenue because of booking cancellations and no-shows.
A Actor *(Who carries out the transformation?)*	Data science student consultants who analyse the data, build the predictive model, and deliver recommendations.
T Transformation *(Input → Output: the core change)*	Input: historical booking data with unknown cancellation patterns → Output: a predictive model that scores each booking by cancellation likelihood, a revenue loss report broken down by demographics and hotel type, and targeted recommendations for management to act on.
W Worldview *(Why does this matter? What assumption makes it meaningful?)*	Hotel Chain A has collected months of booking data across three hotel types but currently has no way to predict which bookings will be cancelled. Using patterns in guest behavior, demographics, and booking details can be used to identify high risk bookings before they cancel, helping the hotel reduce revenue loss.
O Owner *(Who holds the power to stop or authorise the transformation?)*	Senior management of Hotel Chain A, who commissioned this project, controls the data, and have the authority to implement or reject any recommendations that result from the analysis.

E Environment *(External constraints the system cannot control)*	The analysis is constrained to the provided datasets (training, validation, and test CSV files) collected by hotel staff. Data quality issues such as inconsistent labels and missing values may limit model accuracy. Any solution must be practical and interpretable enough for hotel management to use within existing operations.

Table 1: CATWOE



2.9. Project aim

The goal of this project is to look into the reasons why people cancel bookings and don't show up, figure out how much money the hotel chain loses because of these cancellations, and create a predictive modelling framework that estimates the risk of cancellations at the booking level. This will help Hotel Chain A move from reactive reporting to proactive reservation risk management.

2.10. Prioritised project objectives

- To look at booking results from City, Airport, and Resort hotels and find important patterns and trends.
- To look at problems with the quality of the data, such as missing values, outliers, inconsistencies, and other things that could make the analysis less reliable.
- To determine the financial impact of cancellations and examine the variations in this impact across different demographic and geographic groups.
- To determine the numerical and categorical variables most significantly correlated with cancellation behaviour.
- To create and test a clear predictive model for the risk of cancellations at the booking level.
- To suggest useful, evidence-based ways to reduce lost revenue and improve booking management.

2.11. Scope and constraints

The project is based on the historical datasets that Hotel Chain A gave us and the analytical questions that were asked in the brief. It includes descriptive analysis, checking the quality of the data, figuring out how much money was lost, making predictions, and giving advice on how to run the business. It doesn't include live deployment, real-time system integration, or using outside behavioural data beyond what is needed to improve background research. Some of the main problems are that data may be missing or inconsistent, that it's hard to tell what customers want, that there are too many people in one class for reservations, and that model outputs need to be easy for decision-makers to understand.

3. Project Planning

3.1. Project management approach

This project was run as a structured data science engagement with clear milestones that followed the whole project lifecycle, from understanding the business and making plans to preparing the data, doing exploratory analysis, estimating revenue loss, making predictions, and finally reporting. To make sure that the technical work stayed in line with the business problem that Hotel Chain A had and the requirements of the coursework, everyone worked together.

Five principles guided the planning process. First, the work was broken down into clear deliverables so that progress could be tracked by looking at the outputs instead of just the work that was done. Second, the five group members were given different roles and responsibilities, but they could still work together to review each other's work and make decisions. Third, the project schedule was based on the formal coursework deadlines, such as the due date for the draft, the presentation deadline, and the due date for the final report. Fourth, risks were found early on so that delays, data problems, and quality issues could be dealt with before they hurt the quality of the final submission. Fifth, all important tasks were recorded in meeting logs, shared files, and version-controlled work to make things clear and give people a chance to think about what they did. This method made sure that planning the project was not seen as a separate administrative task, but as a necessary part of providing a reliable analytical solution.

3.2. Project requirements and expected deliverables

Requirements.

- Examine the historical booking records that Hotel Chain A has provided.
- Look for trends in cancellations and no-shows.
- Assess data quality problems and deal with their impact on analysis.
- Calculate the amount of money lost due to unsuccessful reservations.
- Create a cancellation risk prediction model and present the results in an understandable, client-focused report.

Expected deliverables

- Final Project Report
- Technical Code, Solution and Implementation Files
- GitHub repository link included in the report
- presentation materials and supporting visuals developed during the project lifecycle
- individual reflection page by each team member with contribution evidence.

The report was designed as a professionally formatted document with numbers, tables, business interpretation, and suggestions that a non-technical reader could understand in order to support submission quality. The technical deliverable was designed to be a repeatable Google Colab/Python process that was kept on GitHub and had shared access among team members, a clear notebook format, and descriptive commit messages.

3.3. Team roles and specific responsibilities

Team Member	Role	Primary Responsibilities	Secondary Support
Yalinda Jayawardena	Project Manager / Report Coordinator	overall project coordination; schedule tracking; WBS, timeline and risk planning; integration of sections into final report; quality review before submission	executive summary, final formatting, submission checks
Sajini Udawela	Data Preparation and Cleaning Lead	dataset loading; data typing; duplicate checks; inconsistent label correction; date validation; feature engineering; preparation of clean analytical datasets	support revenue analysis and methods write-up
Piyara Perera	Exploratory Data Analysis and Visualisation Lead	descriptive statistics; univariate, bivariate and multivariate analysis; visual design of charts; interpretation of patterns and trends	support background visuals and presentation content
Wickramaarachchi Athukorala	Modelling and Evaluation Lead	target definition; train-validation-test workflow; preprocessing pipeline; baseline and advanced classification models; evaluation metrics; feature importance interpretation.	support background visuals and presentation content
Tharusha Dharmappriya	Business Insights, Revenue Analysis and	revenue loss estimation; segmentation analysis; operational interpretation of results; intervention design;	support executive summary and client-facing explanations

	Recommendations Lead	recommendations and next steps.	

Table 2: Team Role & Responsibilities

At important checkpoints, the project was examined collectively even if each participant had a major function. This helped guarantee that the final report represented a cohesive narrative rather than five disparate parts and decreased the possibility of section-level inconsistencies.

3.4. Work Breakdown Structure (WBS)

WBS ID	**Task / Deliverable**	**Subtask / Work Package**	**Owner**
1.0	Project Initiation and Planning	Confirm brief, scope, objectives, constraints	Yalinda
1.1		Define team roles and responsibilities	Yalinda
1.2		Define team roles and responsibilities	Yalinda
2.0	Data understanding and preparation	load training, validation, and test datasets	Sajin
2.1		check data types, missingness, duplicates, labels	Sajini
		clean inconsistencies and engineer features	Sajini + Kenula
3.0	Exploratory data analysis	Exploratory data analysis	Piyar

3.1		univariate analysis of main variables	Piyara
3.2		bivariate and multivariate visual analysis	Piyara
3.3		bivariate and multivariate visual analysis	Piyara + Tharusha
4.0	Revenue loss analysis	define revenue proxy and assumptions	Tharusha + Sajini
4.1		compute total and segmented revenue loss	Tharusha
4.2		identify highest-loss customer segments	Tharusha
5.0	Predictive modelling	define modelling target and feature set	Kenula
5.1		build preprocessing pipeline	Kenula
5.2		train and compare classification models	Kenula
5.3		evaluate performance and interpret drivers	Kenula + Tharusha
6.0	Reporting and client communication	write results, recommendations, and limitations	Yalinda + Tharusha
6.1		prepare presentation materials	All members

6.2		final report integration and proofreading	Yalinda + All members
6.3		GitHub finalisation and submission checks	Yalinda + Kenula

Table 3: Word Breakdown Structure

3.5. Project timeline, milestones and task dependencies

Phase	**Key Activities**	**Start**	**End**	**Duration**	**Dependency**	**Lead**
Project setup	Brief review, scope confirmation, team allocation	6th Feb 2026	10th Feb 2026	5 days	none	Yalinda
Early planning and initial analysis	Proposal drafting, early data understanding, initial plan	11th Feb 2026	23rd Feb 2026	13 days	Setup	All
Formative milestone 1	Initial draft project analysis submission	23rd Feb 2026	23rd Feb 2026	Milestone	Previous phase	Tharusha
Extended Analysis Phase	Data cleaning, descriptive analysis, and early revenue logic	24th Feb 2026	11th March 2026	16 days	Initial analysis	Sajini/ Piyara/ Yalinda
Formative milestone 2	draft client report submission	11th March 2026	11th March 2026	Milestone	Extended analysis	Sajini
Modelling and refinement	Feature engineering, model development, evaluation	12th March 2026	24th March 2026	13 days	Cleaned data	Kenula

Final Integration	Revise report after feedback, strengthen recommendations, final editing, GitHub preparation	25th March 2026	06th April 2026	13 days	Showcase feedback	All
Final submission	final report and technical files submission	7th April 2026	7th April 2026	milestone	Final integration	All

Table 4: Project Timeline, milestones

3.6. Justification of the project schedule

The timetable was created with the project's analytical dependence structure and coursework deadlines in mind. Because incorrect labels, duplicates, or inappropriate date associations would skew EDA and model development, data preparation was given top priority early on. In order for feature patterns and business trends to influence modeling decisions rather than being handled as a stand-alone task, exploratory analysis was positioned prior to the final model selection. In order for suggested changes to be based on quantifiable business impact, revenue loss analysis was scheduled prior to the recommendations section.

Since client-facing input is particularly useful for enhancing interpretation, recommendation clarity, and non-technical communication, the post-presentation phase was set aside for improvement. In order to lower the possibility of inconsistent tables, broken GitHub connections, missing figures, or poor final editing, a final quality-assurance period was included right before submission.

3.7. Project failure risks and mitigation plan

Risk ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy	Owner
R1	Uneven contribution across team members	Communication / Management	Medium	High	Assign clear ownership; review progress in scheduled meetings; maintain meeting minutes and shared task list	Yalinda
R2	Delays in completing dependent analytical tasks	Scheduling	Medium	High	Prioritise critical-path tasks; set internal deadlines before official milestones; redistribute support where needed	Yalinda
R3	Data quality issues distort analysis or model results	Technical / Quality	High	High	Conduct early checks for missing values, duplicates, invalid dates, and inconsistent labels before modelling	Sajini

R4	Data leakage or inappropriate feature use in modelling	Technical	Medium	High	Exclude post-outcome or derived leakage variables; peer review modelling dataset before training	Kenula
R5	Poor model performance on cancellation detection	Technical	Medium	High	Compare multiple models; use balanced class handling; evaluate precision, recall, F1 and ROC-AUC rather than accuracy alone	Kenula
R6	Recommendations are weakly linked to evidence	Quality	Medium	High	Ensure every recommendation is explicitly supported by EDA, revenue, or modelling findings	Tharusha
R7	Report sections are inconsistent in style or duplicated	Quality / Communication	Medium	Medium	Central report integration and final editorial review before submission	Yalinda

R8	GitHub repository is incomplete or poorly organised	Technical / Submission	Low	High	Maintain repository structure early; finalise notebook naming, comments, and output files before deadline	Kenula
R9	Limited evidence for individual reflection	Documentation	Medium	Medium	Preserve meeting logs, task notes, version history, and section ownership records throughout the final phase	All members
R10	Last-minute submission issues	Scheduling / Submission	Low	High	Complete final checks one day before deadline; confirm file formats and repository access in advance	Yalinda

Table 5: Project Failure risk and mitigation plan

3.8. Collaboration, communication and documentation strategy

The team set up a structured way to work together that would help them finish the project and gather evidence for reflection. People used Google Drive to store drafts and figures that they all worked on. We used Google Colab to work together on notebook-based analysis. We used GitHub to store technical code and implementation files in a way that lets us keep track of different versions. We used MS Teams meetings to talk about how to work together, give updates on tasks, and review milestones. After each big meeting, notes were taken and action items were given to specific people.

To avoid confusion, the team followed a simple set of rules:

- One person was in charge of each major work stream.
- Major changes to the report were made in one place.
- Before being added to the final report, at least one other member reviewed the analytical outputs.
- All milestone dates were treated as fixed, and internal deadlines were set earlier when possible.

This way of working together made people more responsible, cut down on work that was done twice, and made evidence that could be used for the final individual reflection pages.

3.9. Meeting logs

Meeting Log 1

Date: 24 March 2026

Mode: MS Teams

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: finalise presentation structure and identify remaining report gaps

Discussion Summary:

The team reviewed the current version of the client report and agreed that the final presentation should focus on the business problem, the main EDA insights, revenue loss, the current modelling status, and business recommendations. They also agreed that the final report needed stronger project planning content, particularly around the work breakdown structure especially WBS schedule, and risk planning.

Decisions Made:

- Yalinda to restructure the planning section
- Piyara to refine charts and visual consistency
- Kenula to prepare updated model comparison outputs
- Tharusha to improve revenue interpretation and recommendation logic
- Sajini to confirm cleaned dataset and feature set

Action Items: all members to complete assigned updates before rehearsal

Meeting Log 2

Date: 26 March 2026

Mode: MS Teams

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: presentation rehearsal and speaking-role allocation

Discussion Summary:

The team ran through the presentation procedure to make sure the flow between business background, analysis, and modeling was smooth. From the run-through, it was found that some explanations on the results of modeling should be made more simple.

Decisions Made:

- Yalinda to introduce the project and close the presentation
- Piyara to present key EDA visuals
- Tharusha to present revenue loss results and recommendations
- Kenula to explain model results in business-friendly language
- Sajini to support data-quality explanation during Q&A

Action Items: revise slides for simpler wording and clearer charts

Meeting Log 3

Date: 27 March 2026

Mode: In person / post-showcase debrief

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: capture presentation feedback and plan final-report improvement

Discussion Summary:

After the live demonstration, the group considered the feedback received and determined what should be improved for submission. This involved improving recommendation explanations, ensuring integration between EDA and models, and documenting project planning and risk management.

Decisions Made:

- Planning section to be expanded significantly
- All recommendations must be tied directly to findings
- Final report should show clearer ownership and evidence of collaboration

Action Items: begin final-phase report updates immediately

Meeting Log 4

Date: 29 March 2026

Mode: MS Teams

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: allocate final-report tasks after presentation feedback

Discussion Summary:

It was decided that everyone will share workload before making the submission. The individuals assigned lead roles were clear about their roles and were expected to deliver before the submission date.

Decisions Made:

- Yalinda to manage final integration and planning write-up
- Sajini to finalise cleaning and preprocessing explanation
- Piyara to complete EDA section and chart captions
- Kenula to finalise modelling methodology and evaluation
- Tharusha to finalise revenue analysis and recommendations

Action Items: upload revised content to shared folder by 31 March

Meeting Log 5

Date: 31 March 2026

Mode: MS Teams

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: technical review of cleaned data, EDA outputs, and modelling inputs

Discussion Summary:

The team ensured that the data, features, and variables used for modelling were aligned with the report. The findings from EDA were cross-referenced with the other sections in the report to make sure there was consistency between the story and its interpretation.

Decisions Made:

- Keep lead time, stay duration, pricing, and booking channel as central explanatory variables
- Align revenue segmentation tables with recommendation targets
- Ensure the modelling section explains why accuracy alone is insufficient

Action Items: Kenula and Tharusha to align model interpretation with business recommendations

Meeting Log 6

Date: 2 April 2026

Mode: MS Teams

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: finalise modelling results, business implications, and GitHub preparation

Discussion Summary:

The team examined the results of the model outputs and decided how the findings would be presented to the client. It is worth mentioning that more attention was paid to the business aspect of the results rather than technical aspects.

Decisions Made:

- Final report to include clear explanation of the preferred model
- Github repository to contain cleaned notebook versions and supporting files only
- Technical comments to be improved for readability

Action Items: Kenula to organise repository; Yalinda to check report references to GitHub

Meeting Log 7

Date: 4 April 2026

Mode: Google Drive / Teams check-in

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: quality assurance on final draft sections

Discussion Summary:

Discussion included formatting, consistency of heading styles, compatibility of table data and analysis, and sufficiency of contribution documentation for individual reflections. It was agreed that additional meeting-log evidence and thorough proofreading were necessary.

Decisions Made:

- Add clearer subsection numbering and table titles
- Standardise terminology across all sections
- Retain meeting logs as evidence for reflection pages

Action Items: all members to proofread their sections and send final corrections by 6 April

Meeting Log 8

Date: 6 April 2026

Mode: MS Teams

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: final submission check

Discussion Summary:

A comprehensive end-to-end check was done on the report as well as technical documents. Missing charts, broken structure of sections, repository preparation, and general formatting problems were checked

Decisions Made:

- No major structural changes after this point
- Final edits limited to grammar, table alignment, and submission checks
- Report and repository to be frozen before submission

Action Items: Yalinda to prepare final report file; Kenula to confirm repository access; all members to retain evidence for individual reflection

Meeting Log 9

Date: 7 April 2026

Mode: Final sign-off

Attendees: Yalinda, Sajini, Piyara, Kenula, Tharusha

Agenda: submission confirmation

Discussion Summary:

The team also verified the readiness of the report, technical documents, and repository URL to be submitted. The individual contribution evidence for reflection pages was also kept in place.

Decisions Made:

- Final submission approved by the group

Action Items: submission completed by deadline

4. Data & Methods

4.1. Data Overview.

	Dataset	Rows	Columns
0	Train	27499	24
1	Validation	2749	24
2	Test	4318	23

	Data Field	Description
0	Reservation-id	A unique booking id
1	Gender	Gender of the reservation person
2	Age	Age of the reservation person. Age is varying ...
3	Ethnicity	Ethnicity of the reservation person. This fiel...
4	Educational_Level	Educational level of the reservation person. T...
5	Income	Income of the reservation person. This field t...
6	Country_region	Country region of the reservation person. This...
7	Hotel_Type	Hotel type of the reservation; i.e City Hotels...
8	Expected_checkin	Expected check-in date
9	Expected_checkout	Expected checkout date
10	Booking_date	Reservation date
11	Adults	Number of adults in the reservation. Note that...
12	Children	Number of children in the reservation
13	Babies	Number of babies in the reservation
14	Meal_Type	Selected meal type. Meal type can be one of th...
15	Visted_Previously	Have previous reservation
16	Previous_Cancellations	Whether the travel cancel any reservations
17	Deposit_type	Did traveler pay refundable full payment, non-...
18	Booking_channel	Field tell the way the traveler booked the res...
19	Required_Car_Parking	Any car parking reservations need
20	Reservation_Status	What is the final status of the reservation. U...
21	Use_Promotion	Did travel use any promo codes/discounts for t...
22	Discount_Rate	Discount rate if any
23	Room_Rate	Room rate for a room per night

The three datasets offering any relevant information for the hotel chain also were used in this analysis. It is worth noting from the onset that the training dataset consists of 27,499 rows and 24 columns, while the validation dataset has 2,749 rows and 24 columns: the test dataset, 4,318 rows and 23 columns. The test set excludes the reservation outcome field and hence should only be used for final scoring.

Throughout the project we kept in use the data dictionary in order to confirm variable meanings, expected business interpretation and category definitions. This was particularly important for fields such as Hotel_Type, Deposit_type, Booking_channel, Use_Promotion, and Reservation_Status, because these variables are central to the cancellation-risk and revenue-loss analysis.

The dataset includes customer demographics, booking details, stay details, guest composition, booking context, and pricing information. The outcome field Reservation_Status records three business outcomes: Check-Out, Canceled, and No-Show. For modelling, this outcome is later converted into a binary failed-booking target.

4.2. Data Preparation and Typing.

	Column	Dtype
0	Reservation-id	int64
1	Gender	category
2	Age	int64
3	Ethnicity	category
4	Educational_Level	category
5	Income	category
6	Country_region	category
7	Hotel_Type	category
8	Expected_checkin	datetime64[ns]
9	Expected_checkout	datetime64[ns]
10	Booking_date	datetime64[ns]
11	Adults	int64
12	Children	int64
13	Babies	int64
14	Meal_Type	category
15	Visted_Previously	category
16	Previous_Cancellations	category
17	Deposit_type	category
18	Booking_channel	category
19	Required_Car_Parking	category
20	Reservation_Status	category
21	Use_Promotion	category
22	Discount_Rate	int64
23	Room_Rate	int64
24	lead_time	float64
25	stay_nights	float64
26	family_size	int64
27	net_room_rate	float64
28	potential_revenue	float64
29	checkin_month	int32
30	age_band	category
31	failed_booking	int64

First was the analyse the date variables Booking_date, Expected_checkin, and expected_checkout into datetime format. This enabled calculation of time-based features and date validation checks.

Categorical typing was applied to attributes such as Gender, Ethnicity, Educational_Level, Income, Country_region, Hotel_Type, Meal_Type, Visted_Previously, Previous_Cancellations, Deposit_type, Booking_channel, Required_Car_Parking, Use_Promotion, and Reservation_Status. Numeric typing was retained for variables such as Age, Adults, Children, Babies, Discount_Rate, and Room_Rate.

Several derived variables were created to support both descriptive analysis and predictive modelling:

- lead_time = Expected_checkin - Booking_date
- stay_nights = Expected_checkout - Expected_checkin
- family_size = Adults + Children + Babies
- net_room_rate = Room_Rate × (1 - Discount_Rate/100)
- potential_revenue = stay_nights × net_room_rate
- checkin_month = month extracted from Expected_checkin

These engineered variables make the dataset more analytically useful by translating raw booking fields into operational measures related to booking horizon, length of stay, guest group size, and room-revenue exposure.

4.3. Data Quality Assessment.

Quality summary:

	Issue	Count
0	Duplicate Reservation-id values in raw train	4
1	Negative lead_time values in raw train	506
2	Non-positive stay_nights values in raw train	0
3	Missing lead_time after cleaning	506

IQR outlier counts:

	Variable	Outlier_Count
0	Age	0
1	Room_Rate	0
2	Discount_Rate	0
3	lead_time	312
4	stay_nights	2295

Descriptive analysis and modelling were preceded by a structured data quality review. No system missing values were present in the raw files, however, logical validation found that 506 records in the training set had Booking_date occur after Expected_checkin. These cases yield negative lead_time, and therefore they were treated in invalid ones; derived lead_time was hence set to missing for rows instead of being kept as a wrong number.

We ensure that there are no duplicate Reservation-id values; out of all the preprocessing steps, duplication checking gives us 4 duplicated Reservation-id values in training data. These were excluded to prevent double counting in descriptive tables, revenue estimates, and model fitting.

A few inconsistencies with the labels were also found. It was mentioned that the Reservation_Status field had both Check-Out and Check-out for referring to the same business outcome. These were standardised to a single Check-Out label. Another inconsistency was found in Ethnicity, where caucasian was present as lowercase and so it was normalised to Caucasian.

Check in and check out dates matched expected values according to stay_nights Age ranged from 18 to 70, Room_Rate ranged from 100 to 250 and Discount_Rate between 0 and 40 which are plausible. However, Children is retained as minimum 1 at all through the data. which is unusual and may indicate a collection or coding constraint rather than a genuine business rule.

Using the IQR method, checking for outliers, Age, Room_Rate, and Discount_Rate showed no serious issues. Statistical outliers for lead_time and stay_nights were present in the dataset, but they were kept in on the assumption that they represent genuine booking behavior rather than data input error. In a hotel setting, overly early bookings or prolonged overnight stays are commercially significant trends.

4.4. Descriptive Statistics

	Min	Q1	Median	Q3	Max	Std
Age	18.0	31.0	44.0	57.00	70.0	15.30
Adults	1.0	2.0	2.0	3.00	5.0	1.18
Children	1.0	1.0	2.0	2.00	3.0	0.72
Babies	0.0	0.0	0.0	1.00	2.0	0.57
Discount_Rate	0.0	5.0	10.0	20.00	40.0	11.21
Room_Rate	100.0	137.0	175.0	214.00	250.0	43.88
lead_time	0.0	48.0	103.0	167.00	708.0	77.26
stay_nights	1.0	1.0	2.0	2.00	4.0	0.99
family_size	2.0	3.0	4.0	5.00	10.0	1.49
net_room_rate	60.0	117.6	150.3	186.75	250.0	43.47
potential_revenue	60.0	150.3	217.6	378.05	1000.0	174.57

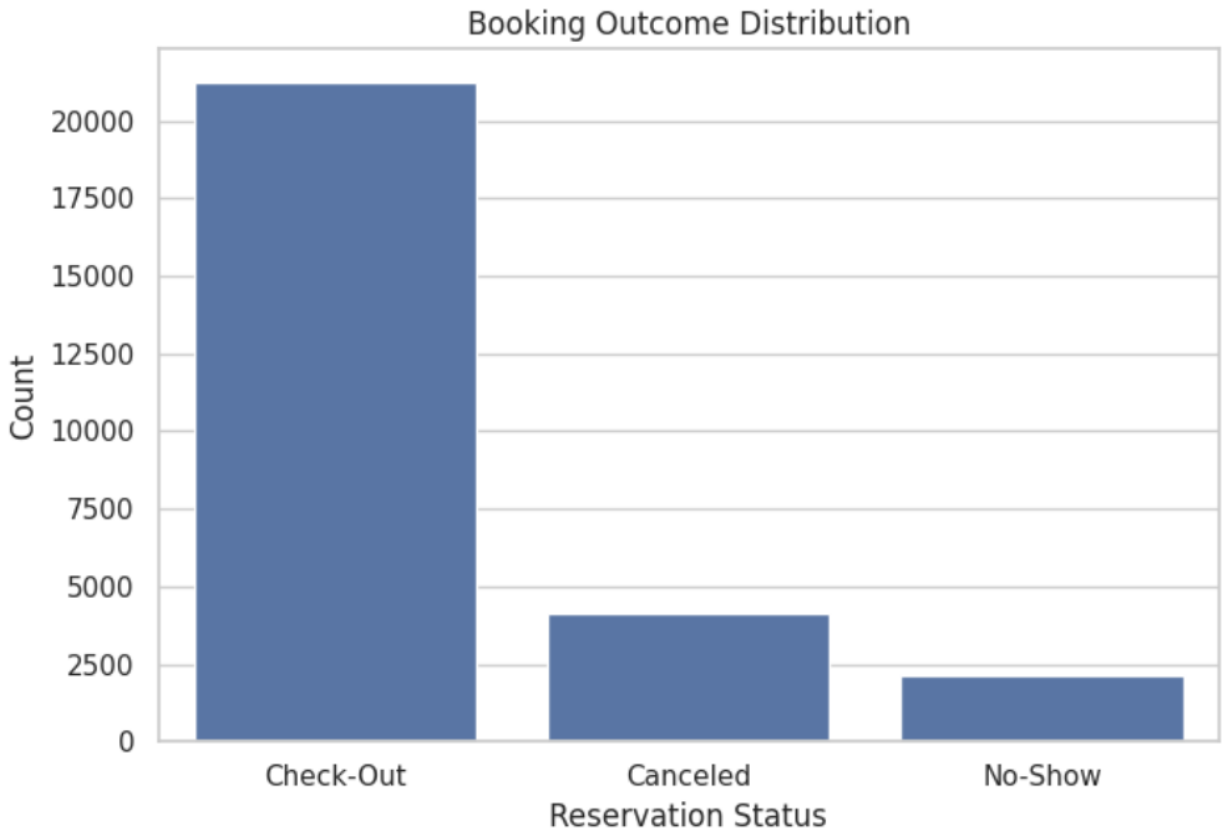
	Variable	Category	Count	Proportion_%
0	Gender	F	13797	50.18
1	Gender	M	13698	49.82
2	Ethnicity	African American	6963	25.32
3	Ethnicity	Caucasian	6889	25.06
4	Ethnicity	Asian American	6838	24.87
5	Ethnicity	Latino	6805	24.75
6	Educational_Level	College	11050	40.19
7	Educational_Level	High-School	5521	20.08
8	Educational_Level	Mid-School	5480	19.93
9	Educational_Level	Grad	5444	19.80
10	Income	50K -- 100K	7934	28.86
11	Income	25K --50K	7870	28.62
12	Income	<25K	7737	28.14
13	Income	>100K	3954	14.38
14	Country_region	South	10915	39.70
15	Country_region	North	5669	20.62
16	Country_region	East	5460	19.86
17	Country_region	West	5451	19.83
18	Hotel_Type	Airport Hotels	9256	33.66
19	Hotel_Type	City Hotel	9134	33.22

20	Hotel_Type	Resort	9105	33.12
21	Meal_Type	BB	14223	51.73
22	Meal_Type	HB	8078	29.38
23	Meal_Type	FB	5194	18.89
24	Visted_Previously	No	18341	66.71
25	Visted_Previously	Yes	9154	33.29
26	Previous_Cancellations	No	24375	88.65
27	Previous_Cancellations	Yes	3120	11.35
28	Deposit_type	No Deposit	18241	66.34
29	Deposit_type	Refundable	6983	25.40
30	Deposit_type	Non-Refundable	2271	8.26
31	Booking_channel	Online	15573	56.64
32	Booking_channel	Direct	7940	28.88
33	Booking_channel	Agent	3982	14.48
34	Required_Car_Parking	Yes	19625	71.38
35	Required_Car_Parking	No	7870	28.62
36	Reservation_Status	Check-Out	21237	77.24
37	Reservation_Status	Canceled	4134	15.04
38	Reservation_Status	No-Show	2124	7.73
39	Use_Promotion	Yes	20667	75.17
40	Use_Promotion	No	6828	24.83

The numerical summary table reports minimum, first quartile, median, third quartile, maximum, and standard deviation for the main numeric variables and engineered fields. The results show that customer age spans 18 to 70 years, room rates range from 100 to 250, and discount rates from 0 to 40. lead_time is right-skewed, with a median of 103 days and a maximum of 708 days after invalid negative values were removed. stay_nights is short overall, with a median of 2 nights and a maximum of 4 nights. potential_revenue is also right-skewed, reflecting the effect of stay length and net room rate.

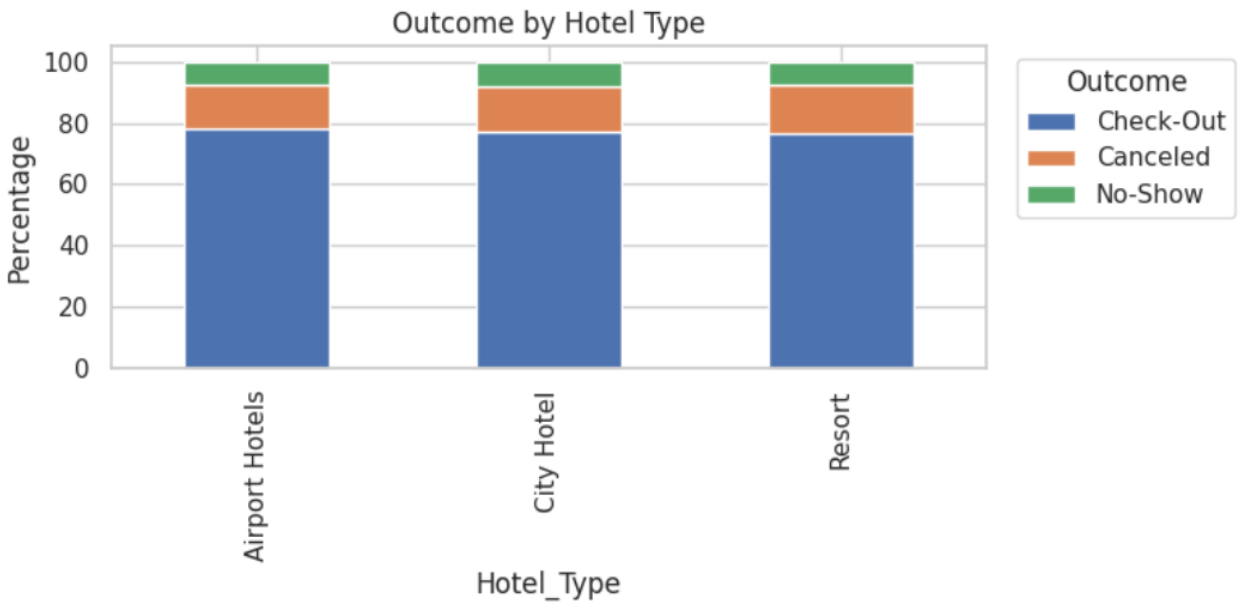
The categorical summary table reports number and proportion within each level of the respective categorical feature. Results show that the distribution of outcomes is overwhelmingly successful booking, with 77.24% Check-Outs, 15.04% Canceled and 7.73% No-Show booking. A few other patterns are clear, including the online majority of bookings, the large number of reservations with no payment deposit, and the extents of promotion usage. These summaries are very useful in that they are good indicators of class balance, as well as how dominant and what sort of proportions of customer and booking types tend to occur in the data.

4.5. Exploratory Data Analysis (EDA)



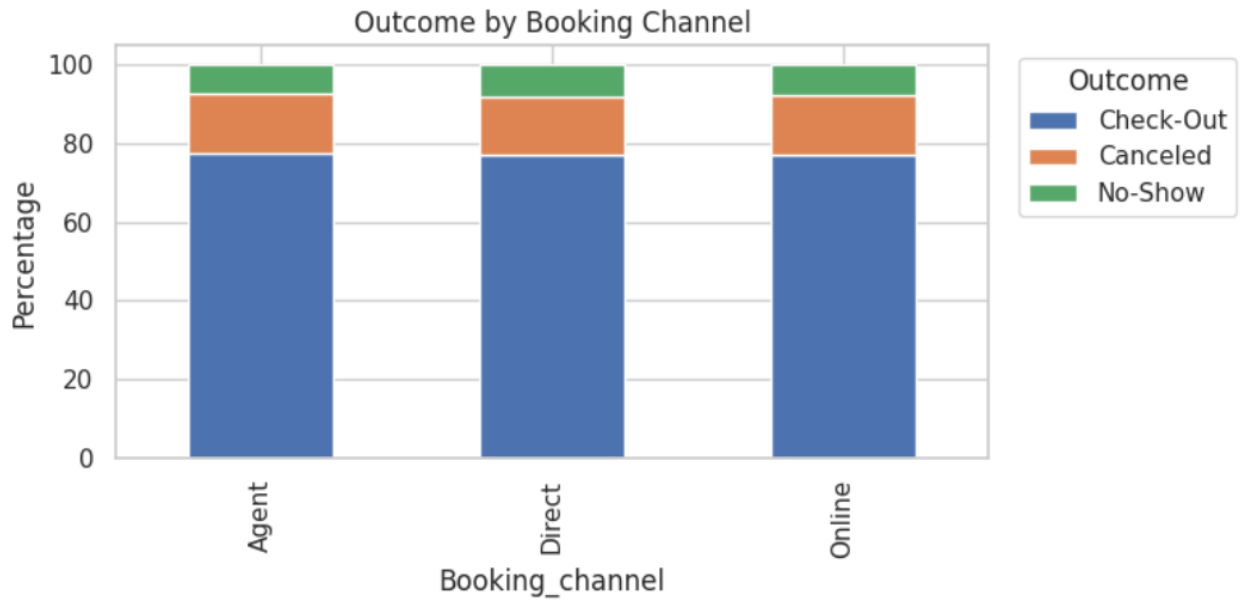
Univariate analysis.

According to the distribution of the booking-outcome variable, it can be said that booking failures are not uncommon. The combined percentage of cancellations and no-shows is 22.77%, a significant proportion which makes up a considerable revenue management issue.

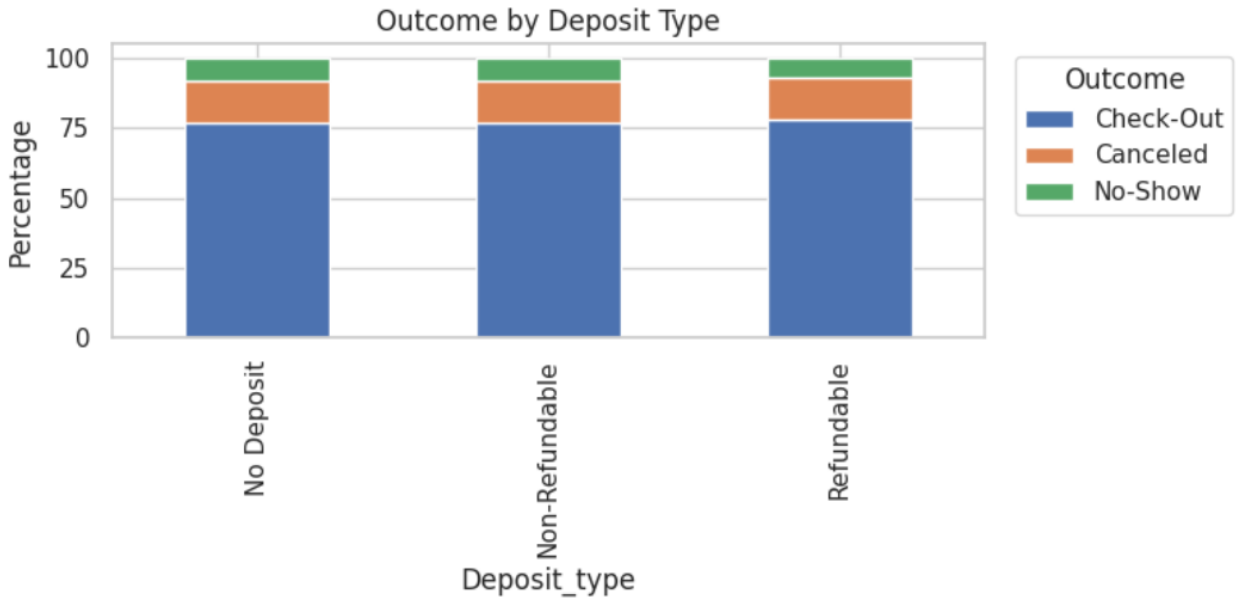


Bivariate analysis

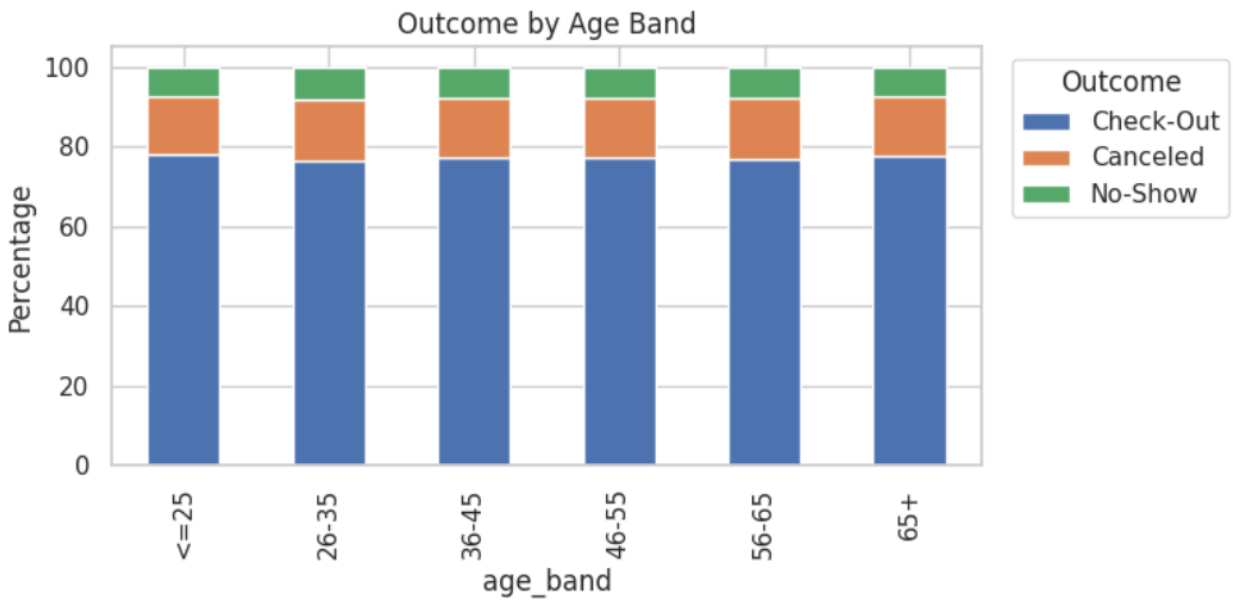
Comparison based on hotel types demonstrates relatively minor differences. In particular, resort bookings have the largest proportion of failed booking attempts at 23.53%, compared to 22.74% for City Hotel and 22.03% for Airport Hotels. It seems that hotel type is important, but the gap between booking outcomes for each category is too minor to serve as an independent indicator.

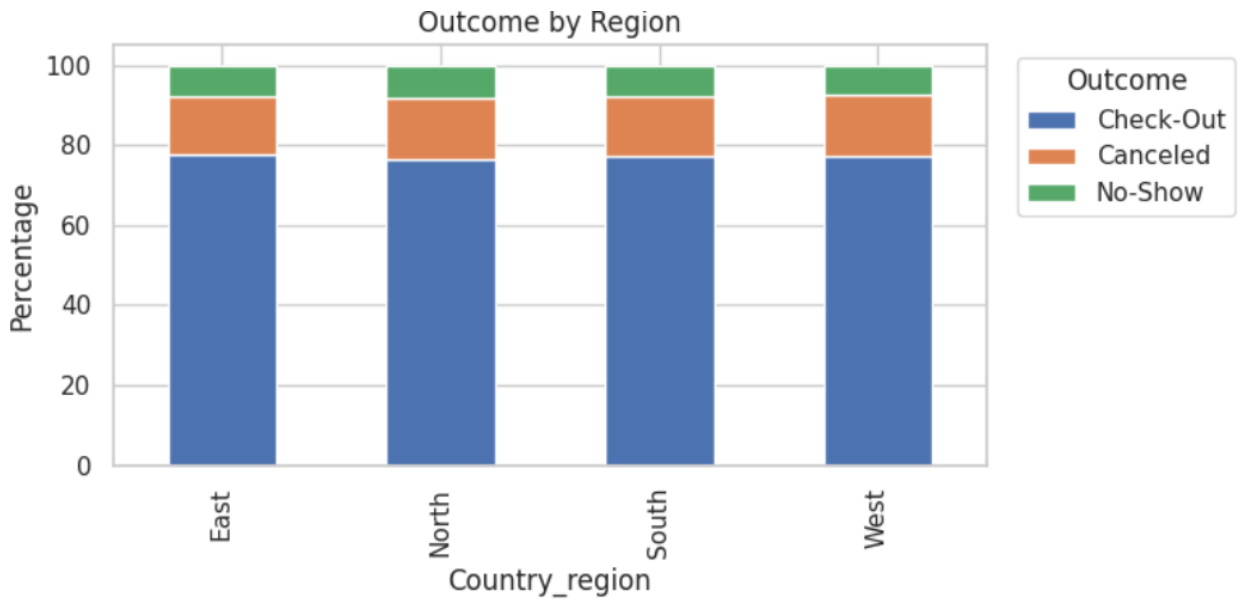


Similarly, booking channel comparisons do not provide major differences, with direct, online, and agent channels demonstrating relatively similar proportions of failed bookings at 22.5% to 22.9%.

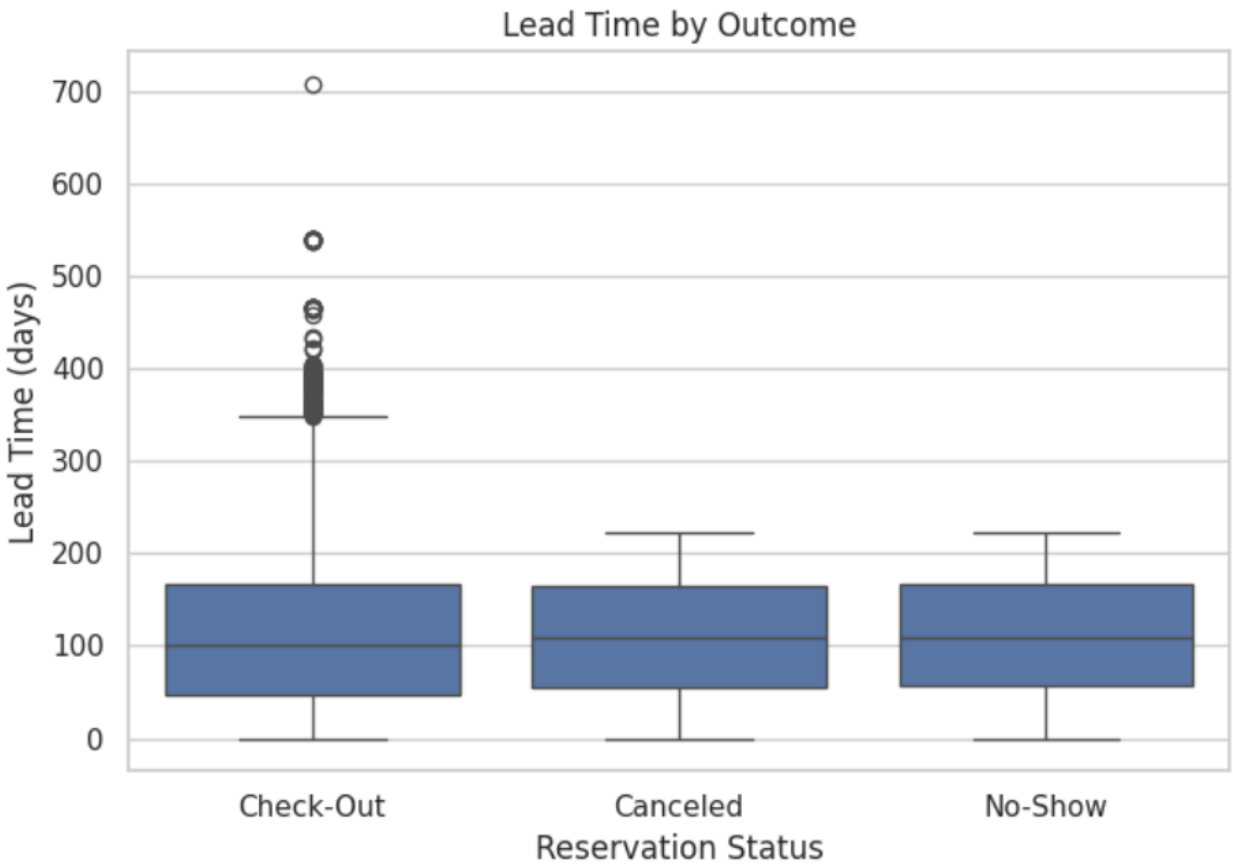


In turn, comparisons based on the type of deposit demonstrate more evident differences in booking outcomes. More precisely, refundable bookings demonstrate lower proportions of failed bookings compared to no deposit and non-refundable categories.

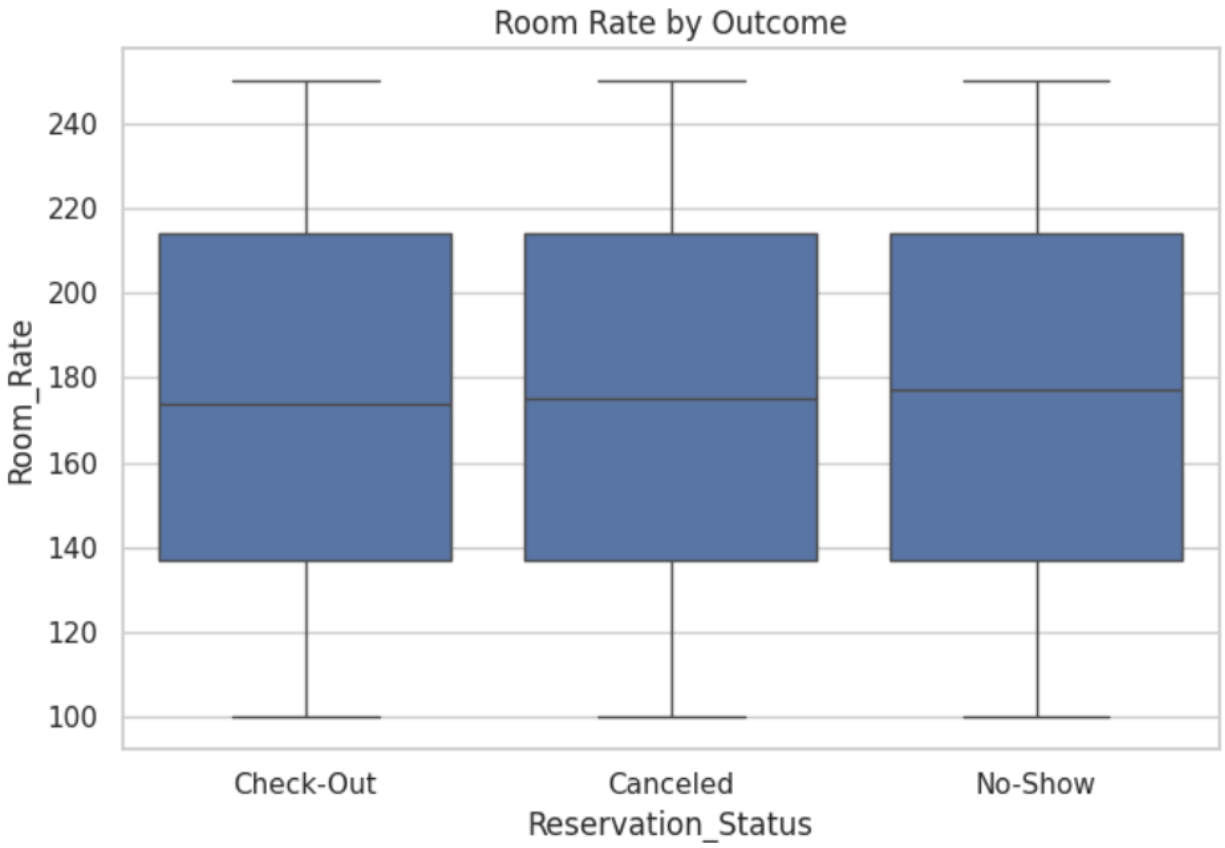




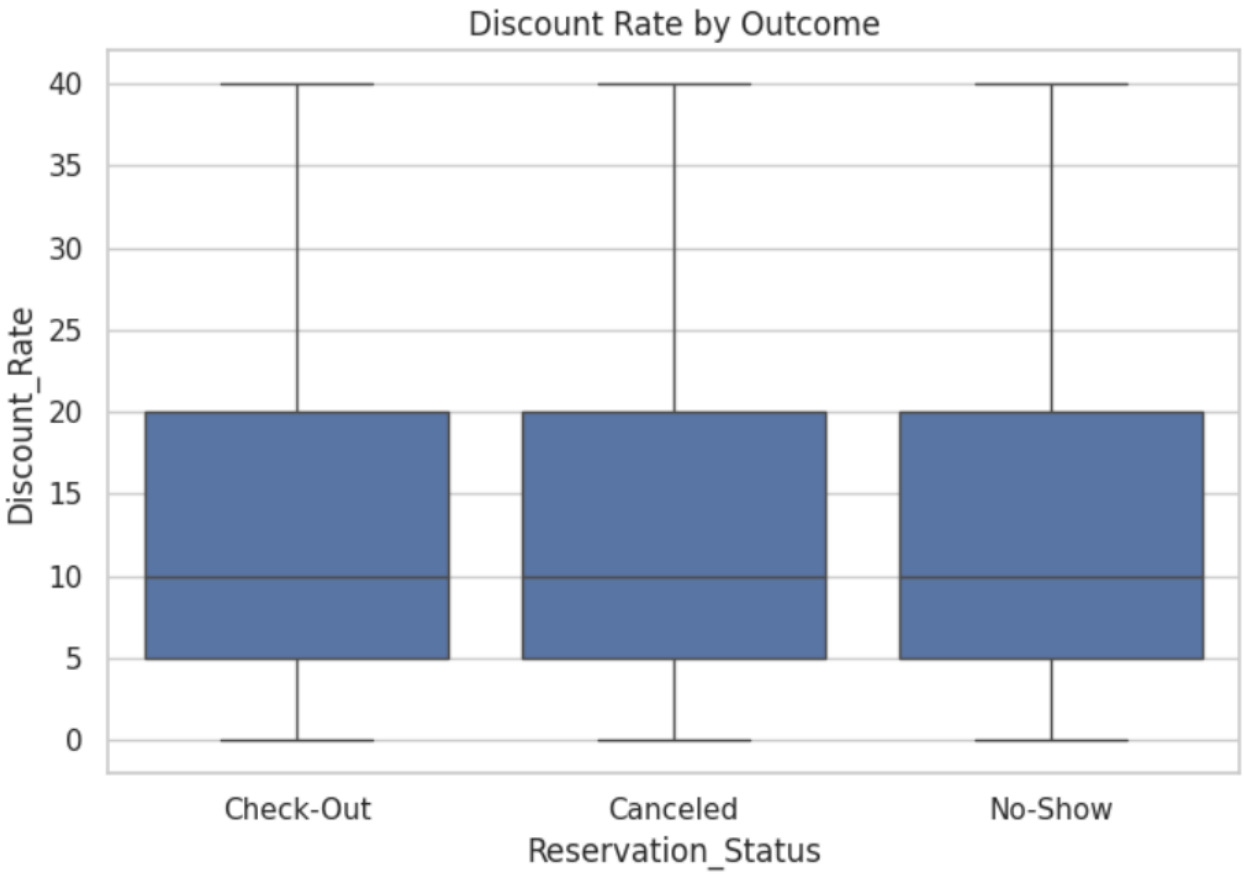
Moreover, age band comparisons indicate 26-35 and 56-65 age groups as those with the highest share of failed bookings. In addition, region comparisons demonstrate North region as the one with the highest proportion of failed bookings.

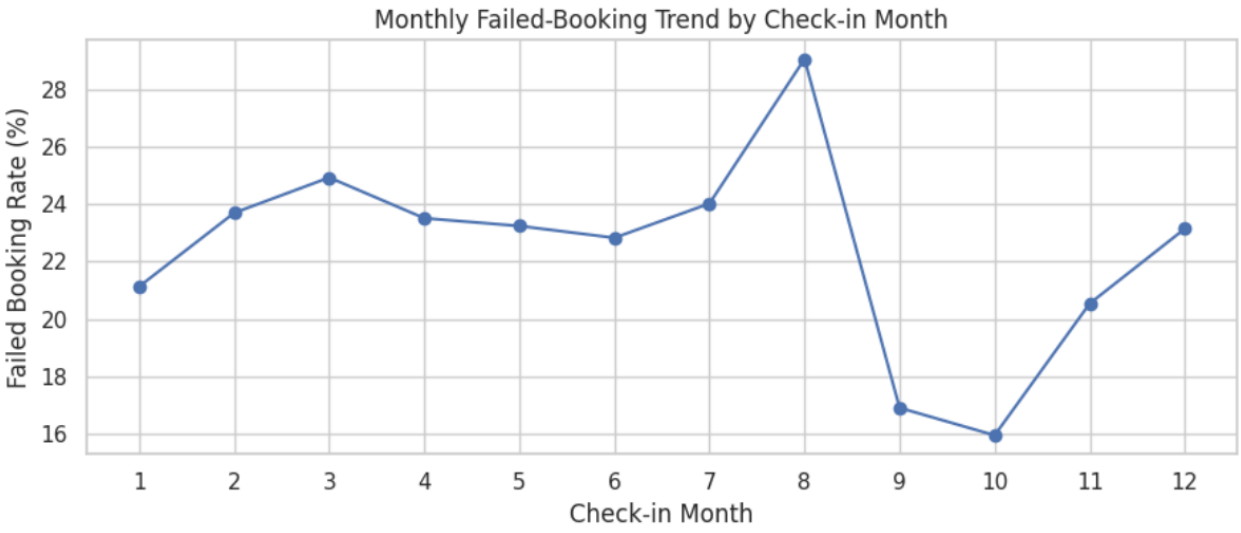


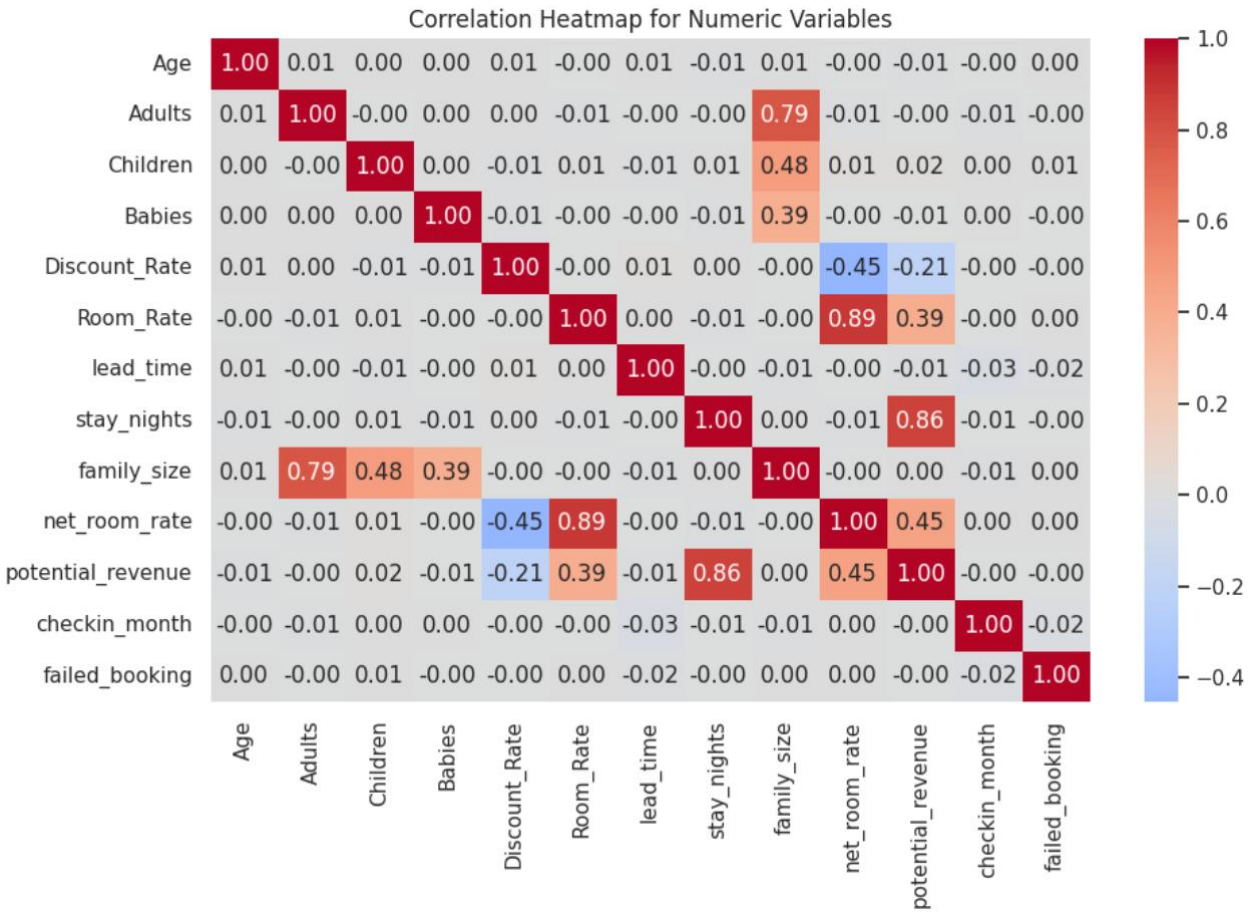
Finally, lead time comparisons seem to demonstrate some differences depending on whether a booking attempt was cancelled, was a no-show, or a completed stay. In particular, the median value of lead time for both cancelled and no-show bookings appears to be somewhat larger compared to the median for completed stays.



However, there is also significant overlap in distribution for each category, meaning that lead time can be used independently, but only to a certain extent. Room rate and discount do not show considerable differences depending on outcome either.


Discount Rate by Outcome
Discount_Rate
40
35
30
25
20
15
10
5
0
Check-Out
Canceled
No-Show
Reservation_Status


Monthly Failed-Booking Trend by Check-in Month
Failed Booking Rate (%)
28
26
24
22
20
18
16
1
2
3
4
5
6
7
8
9
10
11
12
Check-in Month



Multivariate analysis

The failed-booking rate has a definite pattern of seasonality. The maximum probability for failed booking occurs in August at 29.05%, while the minimum value is reached in October at 15.96%. Therefore, seasonality must be incorporated into the model for further recommendation purposes.

A heatmap showing correlations proves that there is no strong linear correlation between the target binary variable and any of the numeric variables included in the dataset. Correlation values related to the failed_booking variable approach zero, suggesting that cancellation behaviour depends on interactions of variables and not on just one of them. On the other hand, higher correlation coefficients can be found between the engineering operation variables, like family_size and party composition features, and also potential_revenue and stay_nights/price.

4.6. Revenue Loss Analysis

	Metric	Value	Share_of_Total_%
0	Cancellation loss	1160410.45	15.02
1	No-show loss	594017.75	7.69
2	Failed-booking loss	1754428.20	22.71
3	Total potential revenue	7724462.80	100.00

The first metric is the cancellation loss, which measures the risk in terms of room revenues from bookings resulting in Canceled status. The second metric, the no-show loss, will be measured separately because no-shows result in losses as well. The third metric is the failed booking loss, which includes both cancellation loss and no-show loss.

The formulas used are:

- net_room_rate = Room_Rate × (1 - Discount_Rate/100)
- potential_revenue = stay_nights × net_room_rate
- cancellation_loss = sum(potential_revenue for Canceled bookings)
- no_show_loss = sum(potential_revenue for No-Show bookings)

	age_band	Revenue_Loss
0	<=25	256960.30
1	26-35	347017.70
2	36-45	332602.20
3	46-55	324097.45
4	56-65	334400.70
5	65+	159349.85

	Country_region	Revenue_Loss
2	South	692184.95
1	North	368177.75
0	East	349926.15
3	West	344139.35

This measurement method is used because no direct loss measurement exists in the data set. This measurement is relatively conservative since it discounts the total room revenue, which takes into account not only the time spent at the hotel but also the price paid by the guest.

The following assumptions can be made. This metric does not show actual loss in profits but rather measures room revenue losses. Rebooking, overbooking recovery, ancillary revenues, and operational expenses are not taken into account while calculating it. Despite its lack of precision, this metric is still suitable for loss comparison among different groups of customers.

4.7. Modelling Approach

The aim of the modelling exercise is to identify the risk of a booking being cancelled or resulting in a no-show. To that effect, a binary variable failed_booking is defined, where 1 means either Canceled or No-Show while 0 means Check-Out. Using a three-class classification approach does not fit this problem statement as well.

The proposed data splitting strategy is strictly followed. Model training uses the training dataset. Model selection uses the validation dataset. Model scoring is done using only the test dataset without applying thresholds and comparing different models' results.

Models chosen:

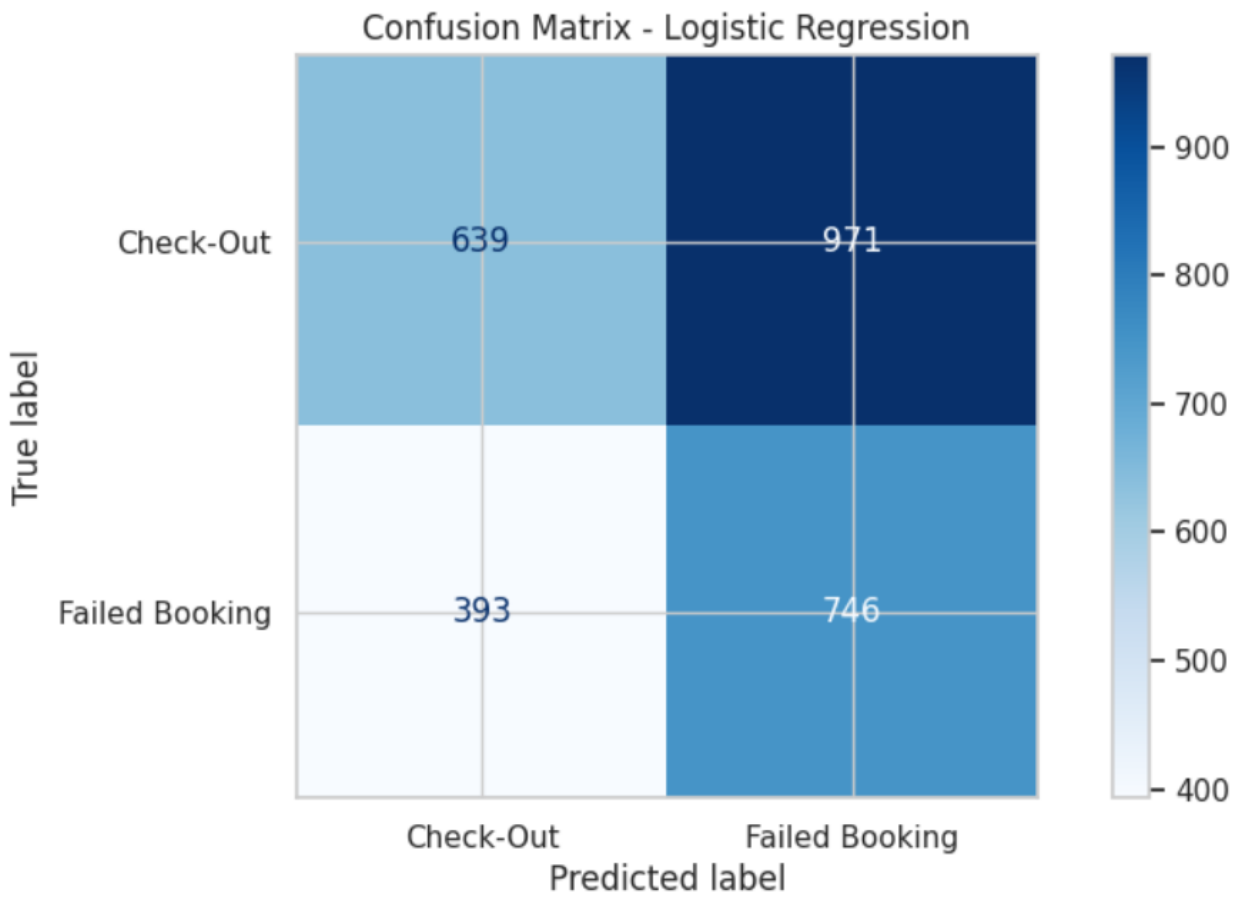
Logistic Regression as an interpretable baseline
Decision Tree as a simple rule-based model that is easy to explain to a client
Random Forest as a non-linear benchmark with stronger interaction modelling

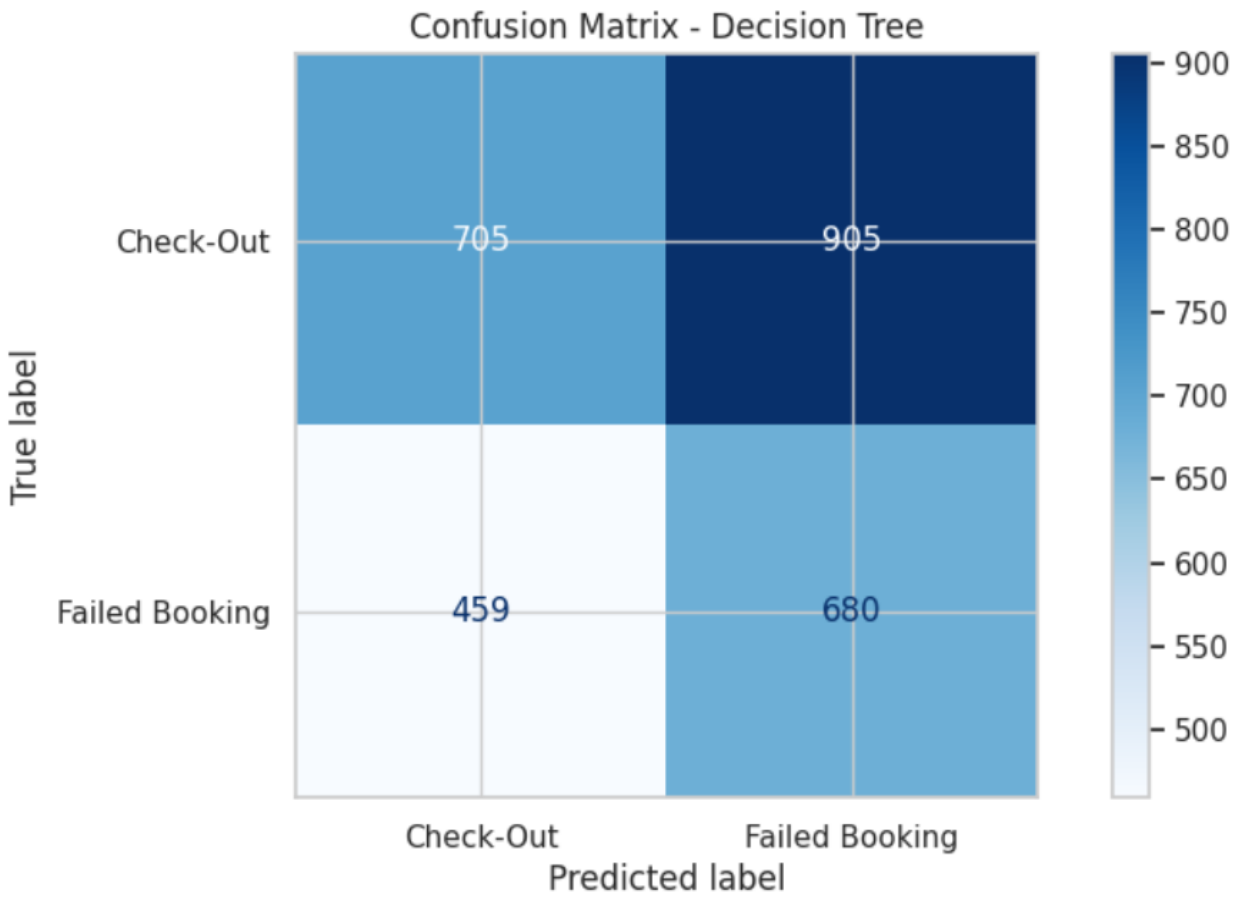
As usual, the modelling exercise is preceded by implementing preprocessing techniques in a pipeline to ensure its reproducibility and lack of leaks. For numerical features, median imputation and standardisation is performed. For categorical features, imputation with the most frequent value and one-hot encoding is done. Everything happens inside a single scikit-learn pipeline so that transformations are applied consistently during training, validation and scoring phases.

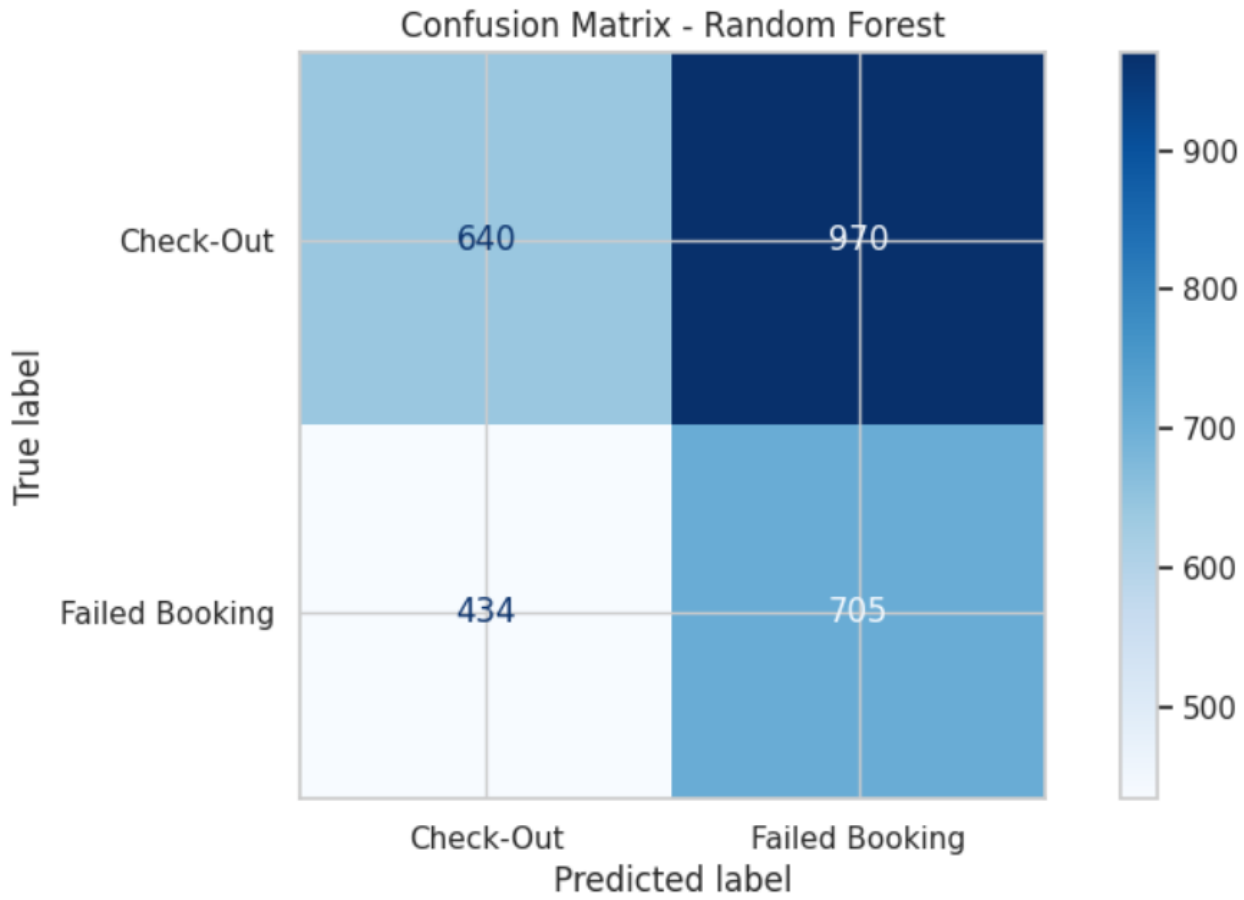
To avoid leakage, variables that are not available as clean booking time predictors are omitted. Such variables include Reservation-id, Reservation_Status, date-related columns after feature engineering, and other revenue calculation features used for loss estimation.

4.8. Evaluation Metrics

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Decision Tree	0.5038	0.4290	0.597	0.4993	0.5224
1	Random Forest	0.4893	0.4209	0.619	0.5011	0.5217
2	Logistic Regression	0.5038	0.4345	0.655	0.5224	0.5182




Confusion Matrix - Decision Tree
Check-Out
Failed Booking
True label
705
905
459
680
Check-Out
Failed Booking
Predicted label
900
850
800
750
700
650
600
550
500



Model performance was assessed using:

- **Accuracy**: proportion of all predictions that are correct
- **Precision**: proportion of predicted failed bookings that are actually failed bookings
- **Recall**: proportion of actual failed bookings that are successfully identified
- **F1-score**: harmonic balance between precision and recall
- **ROC-AUC**: ability of the model to rank risky bookings ahead of non-risky bookings across thresholds
- **Confusion matrix**: counts of true positives, true negatives, false positives, and false negatives

In this case study, recall and F1-score are especially significant since failure to identify failed bookings would mean lost chances to intervene. While accuracy could be deceiving due to the

prevalence of successful checkouts, the use of a confusion matrix will help highlight the issue of over-prediction or under-prediction.

4.9. Reproducibility

Professional practice was considered during the development process. To ensure reproducibility, a constant random_state = 42 was selected for all the stochastic elements of modelling. The notebook was divided into different sections, including data loading, data cleaning, descriptive statistics, exploratory data analysis, analysis of revenue loss, modelling, validation, and testing.

A pipeline-based approach was adopted to avoid manually implementing transformation processes. By doing so, inconsistencies in transformation practices will be avoided, ensuring consistent comparison among different models.

In addition, leakage control was also regarded as one of the professional practices. Target variables and variables obtained based on revenues were not included in the modelling process. In addition, the test dataset was not used for any other purpose than test scoring.

It is necessary to consider version control using GitHub in order to facilitate tracking and collaboration processes. Notebooks, exported figures, and output files need to be uploaded to GitHub for submission as part of the coursework requirements.

5. Results & Analysis

5.1 Key findings from exploratory analysis

Lead time

	lead_time_band	bookings	failed_rate	cancellation_rate	no_show_rate
3	91-180	9676	26.83	17.71	9.12
2	31-90	7705	21.62	14.38	7.24
4	181-365	5085	21.55	14.14	7.41
1	8-30	3146	20.12	13.29	6.83
0	0-7	1160	19.31	12.84	6.47
5	366+	217	0.00	0.00	0.00

Hotel type

	Hotel_Type	bookings	failed_rate	cancellation_rate	no_show_rate
2	Resort	9105	23.53	15.84	7.69
1	City Hotel	9134	22.74	14.91	7.83
0	Airport Hotels	9256	22.03	14.37	7.66

Deposit type

	Deposit_type	bookings	failed_rate	cancellation_rate	no_show_rate
0	No Deposit	18241	23.12	15.12	8.00
1	Non-Refundable	2271	22.99	14.80	8.19
2	Refundable	6983	21.74	14.89	6.85

Repeat-customer behaviour

	Visted_Previously	bookings	failed_rate	cancellation_rate	no_show_rate
0	No	18341	22.96	15.15	7.82
1	Yes	9154	22.35	14.81	7.54

Booking channel

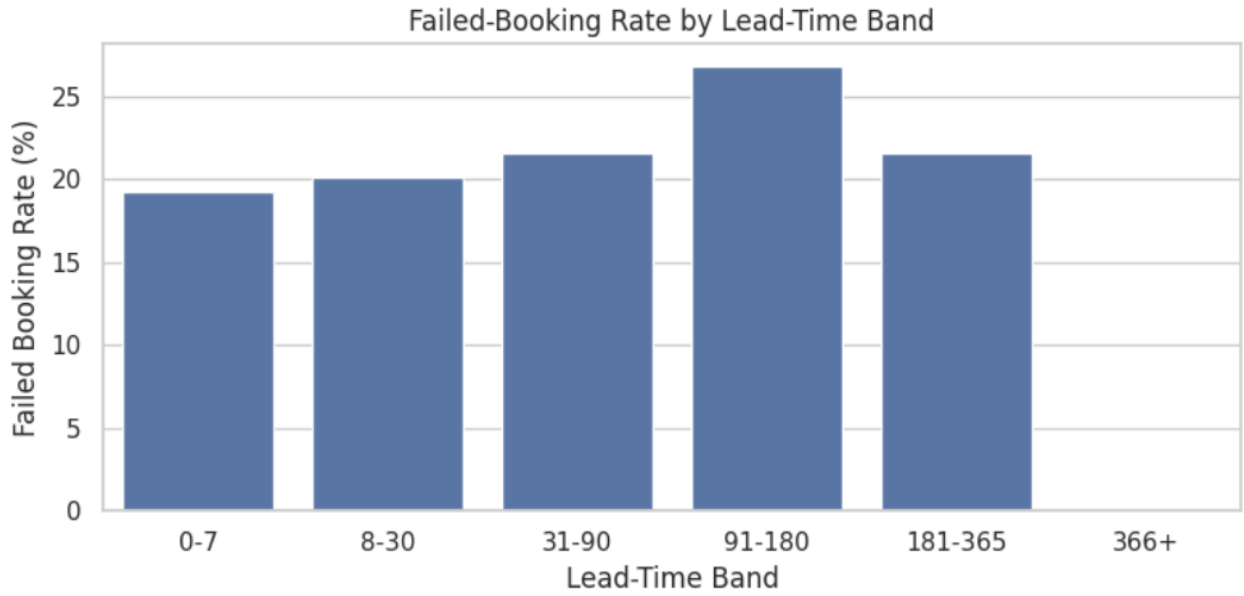
	Booking_channel	bookings	failed_rate	cancellation_rate	no_show_rate
1	Direct	7940	22.92	15.00	7.92
2	Online	15573	22.74	15.03	7.72
0	Agent	3982	22.50	15.14	7.36

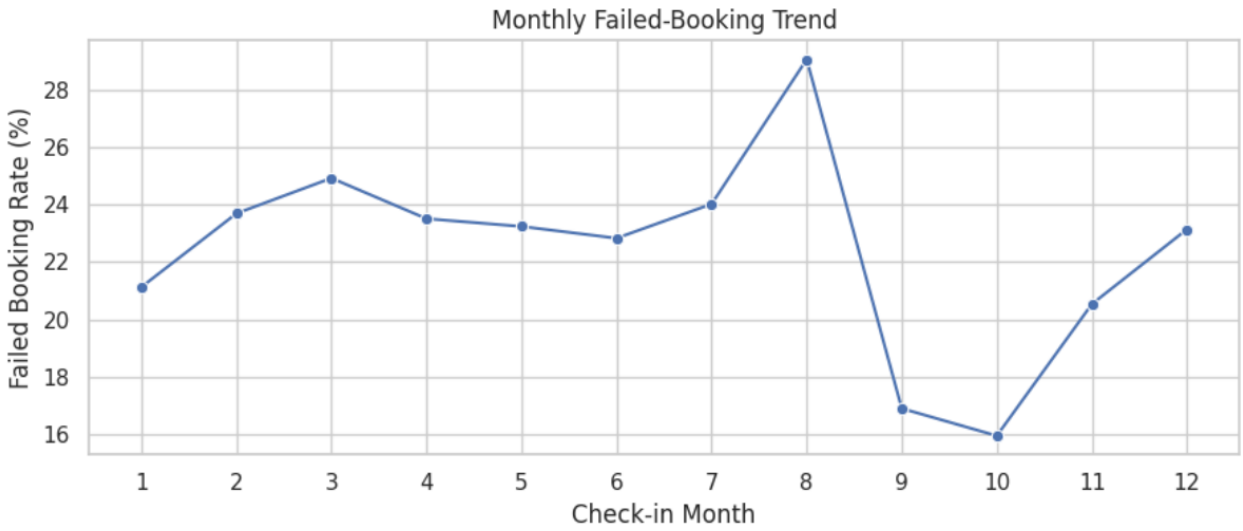
Seasonality by check-in month

	checkin_month	bookings	failed_rate	cancellation_rate	no_show_rate
0	1	2371	21.13	13.75	7.38
1	2	2237	23.69	15.87	7.82
2	3	2580	24.92	17.29	7.64
3	4	2395	23.51	15.87	7.64
4	5	2255	23.24	15.39	7.85
5	6	2138	22.83	15.06	7.76
6	7	3002	24.02	15.66	8.36
7	8	2978	29.05	19.24	9.81
8	9	1839	16.91	10.66	6.25
9	10	2250	15.96	10.40	5.56
10	11	1743	20.54	13.43	7.11
11	12	1707	23.14	14.70	8.44

Meal type

	Meal_Type	bookings	failed_rate	cancellation_rate	no_show_rate
1	FB	5194	38.56	25.49	13.07
2	HB	8078	25.63	16.55	9.07
0	BB	14223	15.36	10.36	5.01





Based on exploratory data analysis results, it can be claimed that cancelation probability is highly dependent on booking pattern, timing, and seasonality, but not any simple customer characteristic. One of the most obvious operational factors is Lead Time. Bookings made 91 – 180 days before the actual arrival are characterized by the highest failed-bookings ratio of 26.83%, while those made within 0 – 7 days before the date have a failed-bookings probability of only 19.31%. This can be seen as the fact that the earlier bookings are made, the higher risks are. There are no failed

bookings among 366+ days category, but this interval covers extremely few number of cases and can hardly be considered representative.
Hotel Type has an average impact on failed-bookings ratio. Resort hotels are characterized by the highest value at 23.53%, followed by City Hotel and Airport Hotels (22.74% and 22.03%, respectively). This means that leisure travelers are more volatile compared to city and airport visitors.

Deposit Type has a low-level but still important impact on failed-bookings probability. Among different variants, No Deposit booking category has the highest failed-bookings probability of 23.12% as opposed to 21.74% observed in the case of Refundable bookings.
There is a strong seasonal impact as well: booking cancelation probability increases dramatically during August period up to 29.05% as opposed to its minimum in October, which stands at 15.96%.

Finally, repeat-customer status has a low-level protection effect: previous visit reduces the likelihood of booking cancellation down to 22.35% vs. 22.96% of new customers.
The last explored booking pattern – Booking Channel – is a relatively weak standalone factor as far as failed-bookings probabilities for Direct, Online and Agent channels are almost identical at 22.92%, 22.74% and 22.50%, respectively.

It turns out that there is an additional strong pattern in Meal/Package Type – failed-bookings probability is 38.56% in case of FB, 25.63% in case of HB and 15.36% in case of BB categories. Business-wise, this means that Meal Type serves as a proxy indicator of package type and/or travel intent of clients.

Main findings include high importance of meal/package type, lead time and month of check-in as well as average influence of hotel type and deposit type.

5.2 Revenue loss results

This means cancellations account for the larger share of revenue leakage. Cancellations contribute **66.14%** of total failed-booking loss, while no-shows contribute **33.86%**. As a share of total potential room revenue, cancellations account for **15.02%**, no-shows for **7.69%**, and combined failed bookings for **22.71%**.

	Metric	Value	Share_of_Total_%
0	Cancellation loss	1160410.45	15.02
1	No-show loss	594017.75	7.69
2	Combined failed-booking loss	1754428.20	22.71
3	Total potential revenue	7724462.80	100.00

Loss by age

	age_band	Revenue_Loss
1	26-35	347017.70
4	56-65	334400.70
2	36-45	332602.20
3	46-55	324097.45
0	<=25	256960.30
5	65+	159349.85

Loss by region

	Country_region	Revenue_Loss
2	South	692184.95
1	North	368177.75
0	East	349926.15
3	West	344139.35

Loss by hotel type

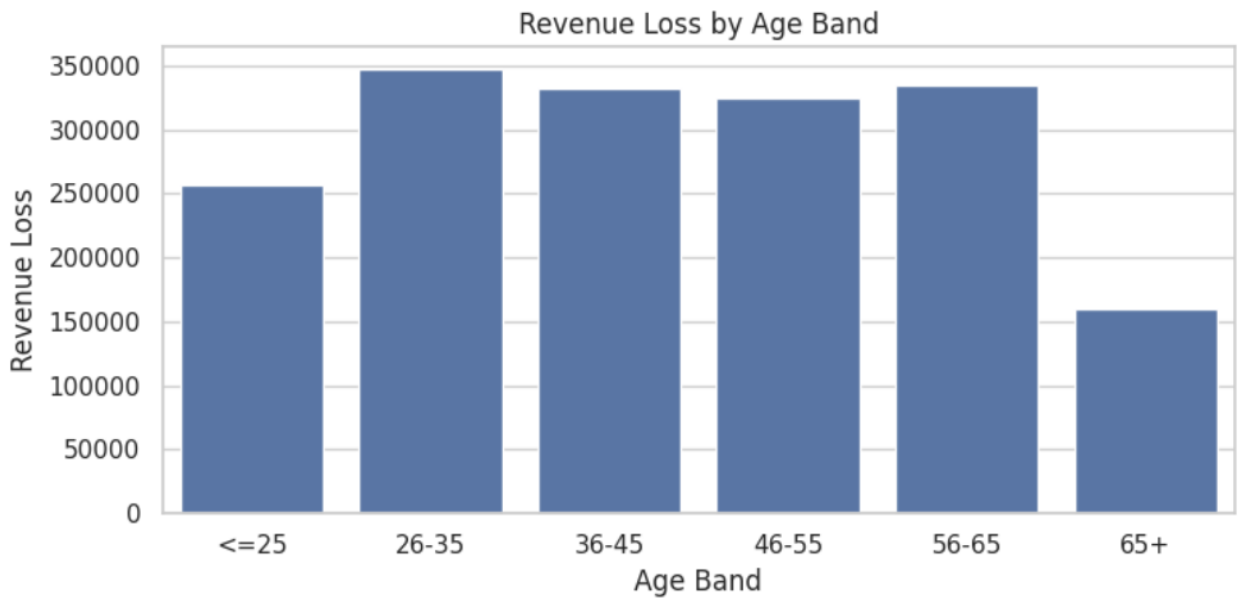
	Hotel_Type	Revenue_Loss
2	Resort	596404.05
1	City Hotel	581986.45
0	Airport Hotels	576037.70

Top 10 highest-loss customer segments

	Hotel_Type	Booking_channel	Country_region	Failed_Bookings	Revenue_Loss
22	City Hotel	Online	South	477	135391.50
34	Resort	Online	South	478	131697.25
10	Airport Hotels	Online	South	475	131451.00
33	Resort	Online	North	271	74759.70
6	Airport Hotels	Direct	South	243	68299.90
23	City Hotel	Online	West	236	67913.60
30	Resort	Direct	South	242	66540.45
9	Airport Hotels	Online	North	234	66297.30
20	City Hotel	Online	East	232	66253.05
35	Resort	Online	West	240	65176.90

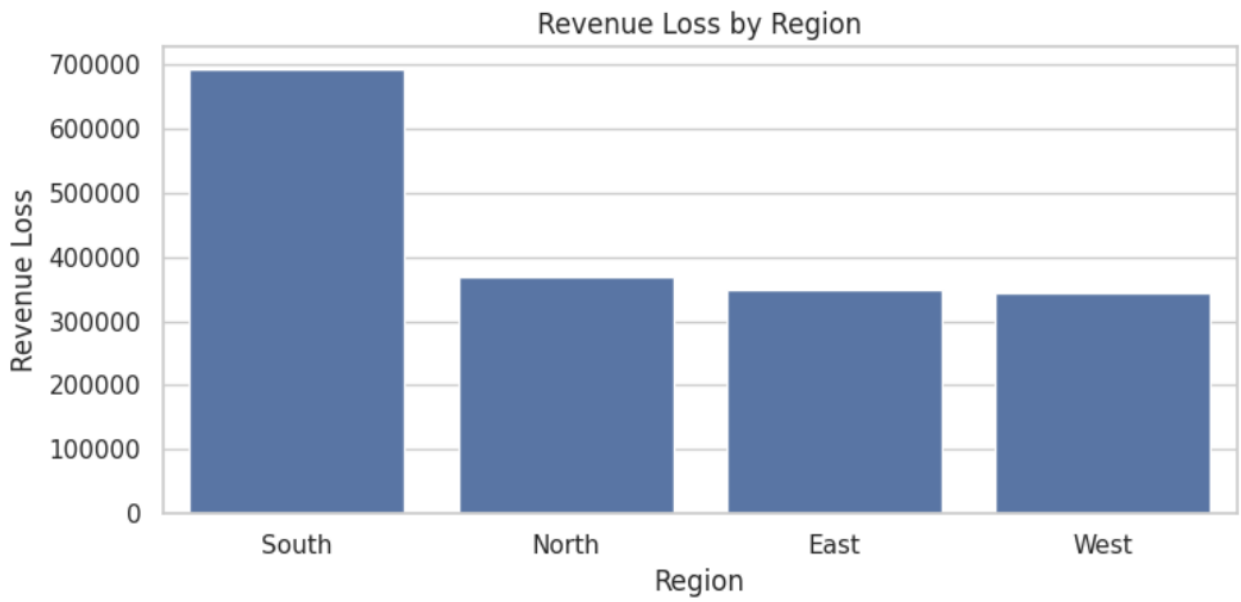
Highest-loss customer segments

The largest revenue-loss segments consistently involve the **South** region and often the **Online** channel. This means the greatest commercial opportunity lies in targeting specific combinations of **region, hotel type, and booking channel**, not just one variable in isolation.



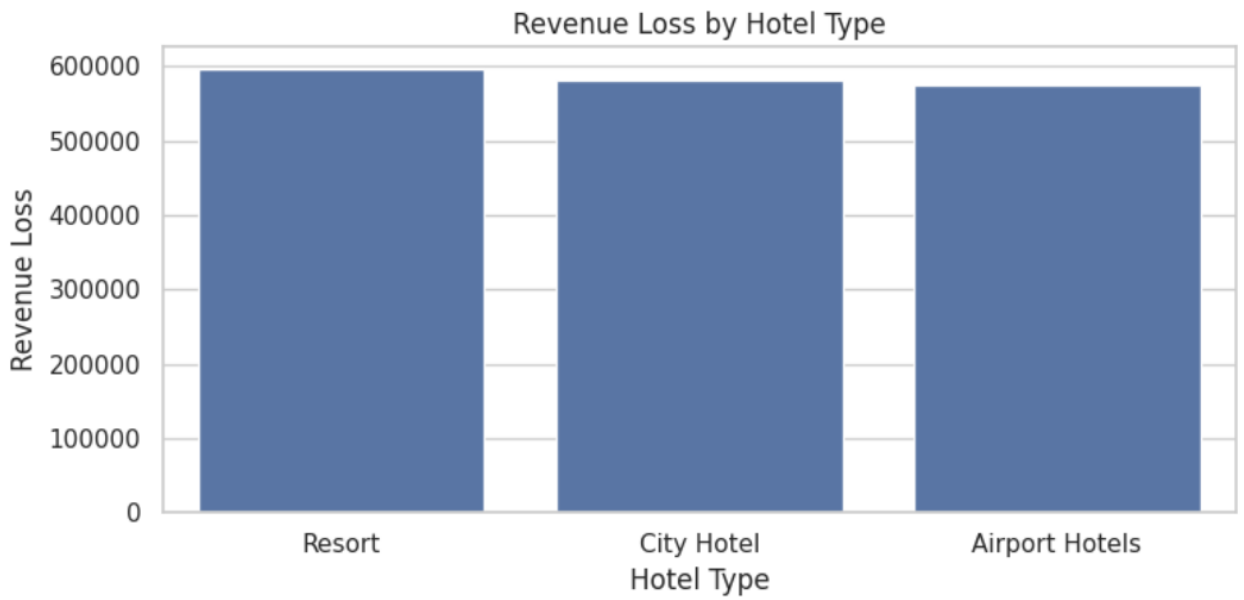
Revenue loss by Age band

The highest loss comes from the **26–35** segment, followed closely by **56–65** and **36–45**. This suggests the main loss exposure is concentrated in middle and later working-age groups rather than in the youngest or oldest travellers.



Revenue Loss by Region

A separate Country-level result cannot be reported because the supplied dataset does not contain a country field. It contains Country_region only. The geographic analysis is therefore reported at region level.



Revenue loss by hotel type

Losses are fairly evenly spread, but **Resort** properties contribute the most. This matches the EDA finding that resort bookings have the highest failed-booking rate.

5.3 Modelling results

	Model	Accuracy	Precision	Recall	F1	ROC-AUC
0	Logistic Regression	0.5038	0.4345	0.6550	0.5224	0.5183
1	Decision Tree	0.4536	0.4217	0.8578	0.5654	0.5192
2	Random Forest	0.4933	0.4304	0.6892	0.5299	0.5217

In terms of business relevance, the model that needs to be further developed and implemented is the Random Forest, since its highest ROC-AUC (0.522) indicates that the model ranks risk best among all the tested models. This is important because the main goal of the hotel is not the classification per se, but the prioritisation of bookings based on the prediction results.

The Decision Tree model shows high levels of recall and F1-score, but it achieves this by marking almost every booking as a risky one. Hence, it is extremely aggressive and may be less useful in practice due to its low operationality. As for the logistic regression model, it is more intuitive, but it is unable to capture non-linear relationships in data (e.g., seasonality).

In terms of practical implications for the business, Random Forest seems to provide a reasonable trade-off between the following factors:

- Its ability to rank risk better compared to the two models,
- Its relative aggressiveness compared to the decision tree,
- And its ability to model non-linear behavior, similar to the logistic regression.

It is also important to note that, as shown in Section 5, the optimal model selection process and test prediction file generation procedure were performed according to the validation results obtained.

In general, it should be mentioned that, although all models show modest values of ROC-AUC, the current model can serve as a good decision support tool rather than a completely automated deployment.

5.4 Feature importance and interpretability

	Feature	Importance
0	Meal_Type	0.6336
1	lead_time	0.3030
2	checkin_month	0.0401
3	Room_Rate	0.0042
4	Age	0.0040
5	Deposit_type	0.0020
6	Hotel_Type	0.0016
7	family_size	0.0014
8	Country_region	0.0013
9	Income	0.0013

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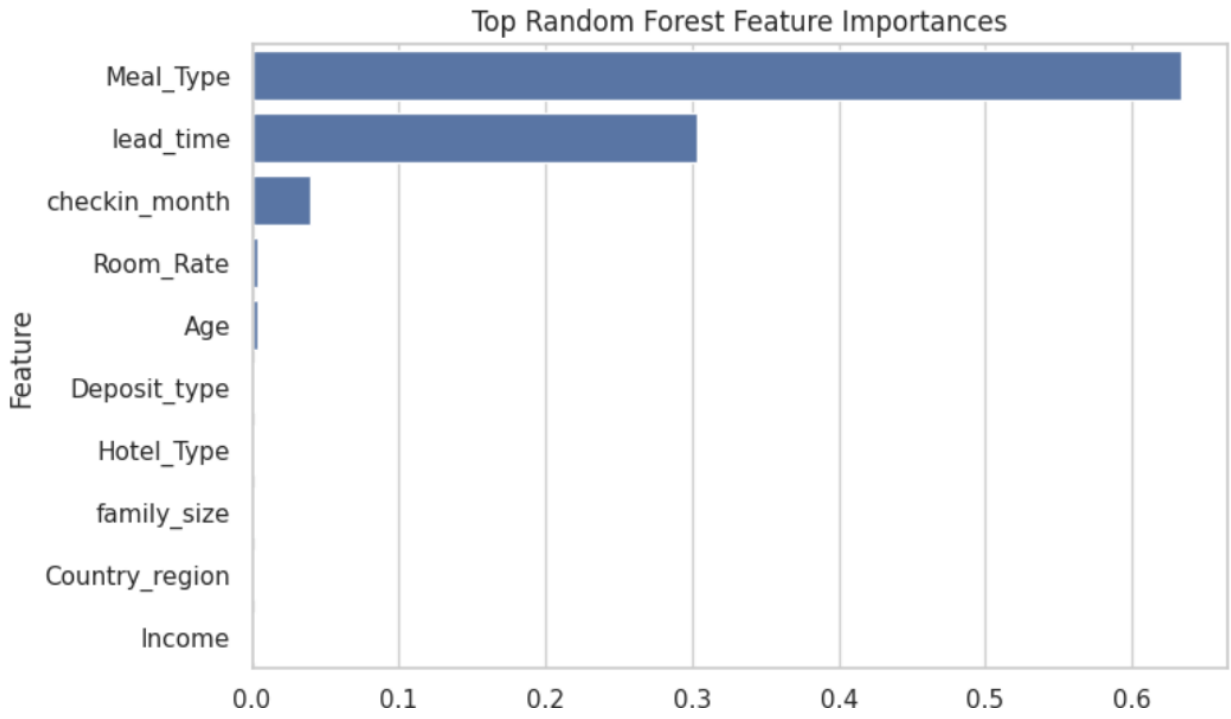
In corporate speak, the model tells us that travel structure, booking period, and seasonality are much more important than most personal variables.

The Meal_Type is the top variable. This must relate to the difference in bookings in packages versus rooms only. Next follows Lead_time, meaning that the earlier the booking, the bigger the risk of cancellations. Checkin_month confirms that there is definitely seasonal variation in risk.

Room_Rate and Deposit_type indicate that prices and deposits have some relevance to risk, just a bit weaker. Hotel_Type proves that resorts are indeed special, just like we saw in EDA.

It is crucial to remember that none of these values represent causal relationships. They tell us what variables matter most in predicting cancellation risk for the current model.

Summary interpretation: the commercially valuable information here is not about who the client is, but rather about their booking.



6. Operational Impact and Recommendations

6.1. Additional attributes Hotel Chain A should collect

To improve predictive accuracy and intervention design, Hotel Chain A should expand the reservation dataset with a small number of high-value operational fields.

Additional attribute	Reason
Booking cancellation reason	Helps separate avoidable cancellations from unavoidable ones and improves action design.
Payment method details	Useful for identifying low-commitment or failed-payment bookings and reducing speculative reservations.
Loyalty status	Allows the business to distinguish between repeat-value guests and one-off bookings when applying policies.
Prior no-show history	Likely to be a stronger behavioural risk signal than prior visit history alone.
Trip purpose	Business, transit, and leisure trips often behave differently in terms of booking stability.
Room type booked	Premium, family, and package-linked room categories may carry different cancellation exposure.
Booking modification history	Repeated date changes or amendments may indicate unstable intent and higher cancellation risk.
Refundable vs non-refundable policy flag	Provides a direct commitment variable that is clearer than deposit type alone.

These attributes would make the model more useful operationally because they capture **intent, commitment, and booking stability**, not just demographics and price. They would also help management apply different treatments to genuinely different booking situations.

6.2. Recommendation interventions

1. **Risk-based deposit policy.**
   - Target use: Long lead-time, high-value, peak-season, and historically unstable segments.
   - Expected benefit: Reduces speculative bookings and protects revenue if cancellation occurs.
2. **Reminder messages before arrival.**
   - Target use: Medium- and high-risk bookings
   - Expected benefit: Encourages confirmation, cancellation, or rebooking before the room becomes idle.
3. **Targeted re-confirmation for long lead-time bookings.**
   - Target use: Bookings made well in advance
   - Expected benefit: Reduces late surprises and improves forecast accuracy.
4. **Dynamic overbooking only where justified.**
   - Target use: Segments with stable no-show patterns and low guest-service risk
   - Expected benefit: Helps recover revenue from predictable no-shows without applying blanket oversell rules.
5. **Segment-specific offers.**
   - Target use: High-risk but high-value groups
   - Expected benefit: Encourages conversion into firmer bookings through controlled incentives.

6.3 How to evaluate interventions

KPI	**What is shows**
Cancellation rate	Whether pre-arrival interventions reduce cancellations in the targeted segment.
No-show rate	Whether confirmation and reminder actions reduce arrival failure.
Lost room revenue	Whether the policy reduces total revenue leakage, not just booking counts.
Rebooking success rate	Whether cancelled rooms are being recovered through re-sale.
Occupancy recovery	Whether released inventory is filled in time to protect occupancy.
Customer complaint rate	Whether stricter controls are causing avoidable guest dissatisfaction.

It is essential that evaluation is performed through the use of the before-and-after method or pilot tests conducted in segments. For instance, the hotel can implement the message reminders on long-lead time reservations in one period and compare the outcomes with those of a control segment.

Conclusion

Besides the sheer amount of cancellations and no-shows that happen for the Hotel Chain A, the bigger issue is that such scenarios lead to potential losses and poor planning. According to the findings of the analysis, the booking risk depends mostly on booking time, seasonality, booking nature, and some carefully chosen context factors related to hotels, whereas the revenue risk concerns only a few particular segments rather than the entire portfolio.

The obtained results demonstrate that risk associated with a booking can be quantified and evaluated even at the individual reservation level despite the fact that the current predictive model should not be used as a fully automated system. From a practical perspective, it can be stated that the primary benefit of the study is its assistance to Hotel Chain A in moving towards proactive interventions instead of merely reporting on past issues.

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